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CO_2 Emissions, Carbon Intensity, and Sustainable Economic Growth in the Western Balkans: A Systematic Literature Review

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Abstract: This study aims to systematically review the literature on CO_2 emissions, carbon intensity, and sustainable economic growth in the Western Balkans, with particular attention to macroeconomic, institutional, financial, industrial, sustainability, and energy-related factors. Following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) methodology, the review was conducted using the Scopus database and structured search strings covering CO_2 emissions, carbon intensity, macroeconomic drivers, energy transition, green finance, urbanization, industrial structure, and sustainable development. After identification, screening, eligibility assessment, and quality assessment, 30 studies were included in the final qualitative synthesis. The findings indicate that CO_2 emissions, carbon intensity, and sustainable economic growth in the Western Balkans are influenced by the interaction among economic growth, energy consumption, fossil-fuel dependence, renewable energy deployment, trade openness, financial development, institutional quality, urbanization, construction activity, carbon pricing, and European Union (EU) climate policy instruments such as the carbon border adjustment mechanism (CBAM). The energy structure emerges as a key determinant of both carbon intensity and CO_2 emissions in the Western Balkans. Continued reliance on coal and inefficient energy systems is associated with higher emissions, whereas renewable-energy deployment and green investment contribute to lower CO_2 emissions and improved carbon efficiency. At the same time, this systematic review identifies important regional differences in fossil-fuel dependence, exposure to carbon-related trade risks, urbanization pressures, construction activity, renewable-energy potential, green investment, and financial-sector transition risks across the Western Balkans. These differences indicate that the region does not follow a uniform decarbonization pathway. The study concludes that effective decarbonization requires an integrated policy framework combining renewable-energy development, stronger institutions, green finance, carbon-pricing mechanisms, sustainable urbanization, and industrial upgrading.

Keywords: Carbon intensity; CO_2; Carbon pricing; Decarbonization; Green transition; Institutional development; Renewable energy

1. Introduction

The twenty-first century is marked by the interconnected challenges of climate change, decarbonization, and sustainable economic development. Increasing trends of carbon intensity cause the environmental degradation, thus underscoring the need for a green transition with a main focus on environmental protection. In this regard, sustainable environmental practices and low-carbon emissions have been gaining increasing attention at a global level. A growing body of research has examined, the determinants of carbon intensity, CO_2 emissions and green practices. Over the past few years, Western Balkan countries, including Kosovo, North Macedonia, Albania, Montenegro, Bosnia and Herzegovina, and Serbia, have been facing growing pressure to accelerate their green transitions and harmonize their environmental and economic policies with those of the European Green Deal and the Green Agenda for the Western Balkans. Even though much progress has been achieved, the countries are still confronted by many structural, institutional, and energy problems related to their high carbon dependency,

reliance on fossil fuels and limited decarbonization progress. The current literature on carbon intensity and sustainable development places growing emphasis on identifying the determinants of carbon intensity at macroeconomic, environmental, technological, and institutional levels. This furthermore implies the need for a green and sustainable growth model that reduces carbon intensity. In this regard, reducing CO_2 emissions and carbon intensity highlights the need for innovation in production systems, investment, labour-market transformation, and technological progress. The World Bank's World Development Indicators report total CO_2 emissions as the aggregate quantity of carbon dioxide emissions, commonly expressed in million tonnes of CO_2 equivalent, while per-capita emissions measure the emissions of an economy relative to its population (World Bank, 2024a). The World Bank further defines the carbon intensity of GDP as the amount of CO_2 emissions produced per unit of an economy's gross domestic product, measured using either constant 2015 US dollars or 2021 purchasing-power-parity GDP (World Bank, 2024b). The carbon intensity of an economy or a production process is a measure of its efficiency of emissions, while the CO_2 emissions show the overall extent of environmental pressures. The decrease in carbon intensity does not always mean the decrease in the total CO_2 emissions, especially in case the economy, production, and energy consumption keep increasing. The total CO_2 emissions can be decreasing simply due to the decrease in the economy's productivity. Therefore, in this systematic review, the environmental performance will be considered as two distinct yet interrelated dimensions of CO_2 emissions and carbon intensity. The European Climate Law makes the European Union's (EU's) commitment to climate neutrality by 2050 binding, establishing a target to reduce greenhouse gas emissions by at least 55% by 2030, with the proposed reduction of 90% by 2040 (European Commission, 2026). Furthermore, the European Green Agenda developed by the EU outlines concrete strategies to support individual countries towards the green transition. The Western Balkans region, remains largely dependent on fossil fuel energy and is characterized by high carbon intensity, mainly due to limited investment in renewable energy, insufficient institutional capacities, and outdated energy infrastructure (European Commission, 2020; Regional Cooperation Council, 2020a). The existing body of literature, reflects broad academic debate on the relationship among economic growth, energy use, carbon intensity, CO_2 emissions climate change, and the transition toward the green economy. These literature evidence present different methodological and empirical approaches, with many studies focusing on developed countries or the EU, attempting to shed light on the impact of foreign direct investment (FDI), international trade, renewable energy, and CO_2 reduction (Sarkodie & Strezov, 2019; Shahbaz et al., 2021).

In addition, Işık et al. (2025a) have revealed that the contribution of the consumption of renewable energy is positive to the achievement of the Sustainable Development Goals, while the insignificant effect of climate policy uncertainty on carbon intensity may reflect insufficient international climate policy coordination (Işık et al., 2025b).

Kasman & Duman (2015) found an inverse U-shaped relationship between environment and income, consistent with the Environmental Kuznets Curve (EKC) hypothesis. While contrary to this finding, Ziberi et al. (2025) found an inverted U relationship between economic growth and environmental degradation in the case of Western Balkan countries. The study concluded that the EKC hypothesis is not universal and depends on the countries stage of development and the econometric conditions under investigation. Furthermore, the evolving global economic and environmental conditions has intensified the need for transition towards a low-carbon economy and requires policy action and substantial investment in cleaner products, production processes, and technologies, particularly in carbon-intensive sectors such as energy and transport (Linnenluecke et al., 2019).

These theoretical approaches indicate that the determinants of carbon emissions vary across countries and that the green transition does not follow a uniform path. Instead, its trajectory and speed depend on each country's economic structure, institutional capacity, and policy framework. Furthermore, it is important to identify the key determinants of CO_2 emissions and carbon intensity in order to design effective policies that reduce environmental degradation while supporting sustainable economic growth. One of the primary concerns during this transition is the complex relationship between environmental sustainability and economic growth, particularly in developing and emerging economies. Economic growth, energy consumption, industrialization, and trade openness are broadly acknowledged as important determinants of CO_2 emissions and carbon intensity. Changes in economic structure, together with technological progress, therefore play a central role in supporting the transition towards a sustainable and low-carbon economy. This perspective is closely aligned with the main aim of this systematic literature review, which examines how economic, energy-related, institutional, and technological factors influence CO_2 emissions and carbon intensity in the Western Balkans. Nevertheless, the majority of research has focused on the relationship between GDP and carbon intensity or CO_2 emissions, frequently neglecting other crucial macroeconomic factors that may be crucial to the low-carbon transition such as: (i) FDI, (ii) trade openness, (iii) government spending, (iv) public debt, (v) labor market conditions, and (vi) the renewable energy consumption. In light of this, it is important to examine how these factors, together with economic growth, influence CO_2 emissions and carbon intensity over time in the Western Balkan countries.

Although the literature on decarbonization and environmental sustainability in the Western Balkans has expanded, the existing review evidence remains fragmented, with most studies focusing on individual sectors rather than providing a broader integrated perspective. Earlier review-oriented studies have focused on specific

dimensions, including the relationship between climate change and agriculture, environmental and development finance, and the decarbonization of the energy sector. While these studies provide valuable insights, still do not provide a comprehensive review of the evidence in question in regard to CO_2 emissions, carbon intensity, and sustainable economic growth in the case of Western Balkans. Different geographical regions, environmental measures, time spans, variables, and methodology used also complicate the formulation of definitive conclusions. In the context of this study, using the Scopus search strings, there were several studies that combine all six of these aspects in a unified framework under the context of the Western Balkans. In an effort to bridge this gap, the present study makes use of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) approach to find, categorize, and synthesize 30 relevant studies under six thematic area. In this systematic review, the distinction is made between absolute CO_2 emissions, CO_2 emissions per capita, and carbon intensity; the similarities and differences in the geographic and methodological settings of the selected studies are analyzed, and, convergence, divergence, and knowledge gaps are determined. Therefore, the main value of this study lies in its systematic review and in recognizing the methodological geographical, and thematic diversity that should be considered in future research.

The main aim of this study is to assess the systematic literature review based on PRISMA guidelines, on CO_2 emissions, carbon intensity, and sustainable economic growth in the case of the Western Balkans from the perspective of sustainable economic growth. Specifically, this study tends to highlight the key macroeconomic, structural, environmental, institutional, and energy-related determinants associated with carbon intensity and CO_2 emissions. Furthermore, the study also tends to analyze how the transition toward renewable energy, industrialization, trade liberalization, urbanization, and sustainability-related policies contribute to reducing carbon intensity and improving environmental outcomes in the long term. Using a systematic literature review methodology based on PRISMA guidelines, this study synthesizes the existing literature, identifies key knowledge gaps, and enhances understanding of the challenges associated with sustainable and low-carbon economic development in the Western Balkans.

1.1 Research Questions

Based on the identified gaps in the literature and the need for a systematic synthesis of the factors influencing the carbon intensity in the Western Balkans, the following research questions guide this study. These questions were developed to examine the key economic, energy-related, institutional, urban, industrial, and sustainable development factors identified in the reviewed literature.

RQ1: What insights does the existing literature provide on CO_2 emissions and carbon intensity in the Western Balkan countries?

RQ2: What macroeconomic determinants, such as economic growth, GDP, financial development, FDI, inflation, and trade openness, are commonly associated with CO_2 emissions in the Western Balkans?

RQ3: What effect do energy structure, fossil fuel dependence, renewable energy use, energy consumption, and energy efficiency have on CO_2 emissions and carbon intensity in the Western Balkans?

RQ4: What role do governance quality institutional quality, environmental regulations, carbon taxation, carbon pricing, the EU Green Agenda, and carbon border adjustment mechanism (CBAM) play in shaping the decarbonization outcomes in the Western Balkans?

RQ5: How do urbanization, industrialization, the construction sector, manufacturing, and value-added industries affect CO_2 emissions and carbon intensity in the Western Balkans?

RQ6: What relationship among CO_2 emissions, carbon intensity, green growth, sustainable development, and decarbonization are identified in the reviewed literature on the Western Balkan countries?

Carbon intensity is affected by economic activity, energy composition, trade openness, consumption patterns, and demographic characteristics.

Figure 1 presents the theoretical conceptual framework employed in this study to examine the determinants of CO_2 emissions, carbon intensity, and sustainable economic growth in the case of the Western Balkans. It draws on key theoretical and policy frameworks, including the EKC, Green Growth Theory, Energy Transition Theory, Structural Change Perspectives, the European Green Deal, and the Green Agenda for the Western Balkans. The six themes surrounding the framework represent the main analytical dimensions of CO_2 emissions and carbon intensity in the Western Balkans. These dimensions encompass macroeconomic factors, energy structure, governance, urbanization, industrial development, and green growth.

The contribution of this study lies in providing a structured thematic synthesis of the existing literature on the factors associated with CO_2 emissions and carbon intensity in the Western Balkans. Rather than examining these factors in isolation, the review considers how six analytical dimensions—macroeconomic conditions, energy systems, governance quality, urbanization, industrialization, and green-transition policies—interact in shaping carbon-related outcomes across the region.

1.2 Existing Evidence on Carbon Emissions, Green Transition, and European Green Deal Alignment

This section covers the theoretical approaches, empirical evidence, and relevant insights regarding the determinants of carbon intensity in the context of the Western Balkan countries. The green economy has been positioned as a framework for balancing economic development, environmental protection, and social well-being within key policy frameworks including: (i) towards a green economy (United Nations Environment Programme, 2011); (ii) green growth (OECD, 2011); (iii) and inclusive green growth (World Bank, 2012). The 2030 Agenda for the Paris Agreement (United Nations, 2015a), Sustainable Development (United Nations, 2015b), and the European Green Deal (European Commission, 2019) together strengthened this agenda by making climate neutrality a key policy objective. Furthermore, the International Labor Organization (2018) highlighted the significance of green jobs and labor markets as important drivers of the green transitions. By examining the definitions, instruments, and mechanisms of the green economy, scholarly research has further developed and complemented these institutional frameworks. Loiseau et al. (2016) examined a variety of approaches, showing how they contribute differently towards to strong and weak sustainability. The green economy improves resilience and mitigates reliance on fossil fuels (Musango et al., 2014). Whereas, Xu et al. (2022) emphasized that emphasized that, while green technology innovation accelerates the low-carbon transition, local government competition may impede progress. Furthermore, Wang (2023) noted that Intergovernmental Panel on Climate Change emphasizes the need for a global response to ensure sustainable development, reduce the risks of poverty and climate change, and stabilize greenhouse gas (GHG) emissions. Several studies have addressed the determinants of the green transition alongside carbon emissions in Western Balkan countries from different perspectives, including the EKC hypothesis (Ziberi et al., 2025) and the green finance.

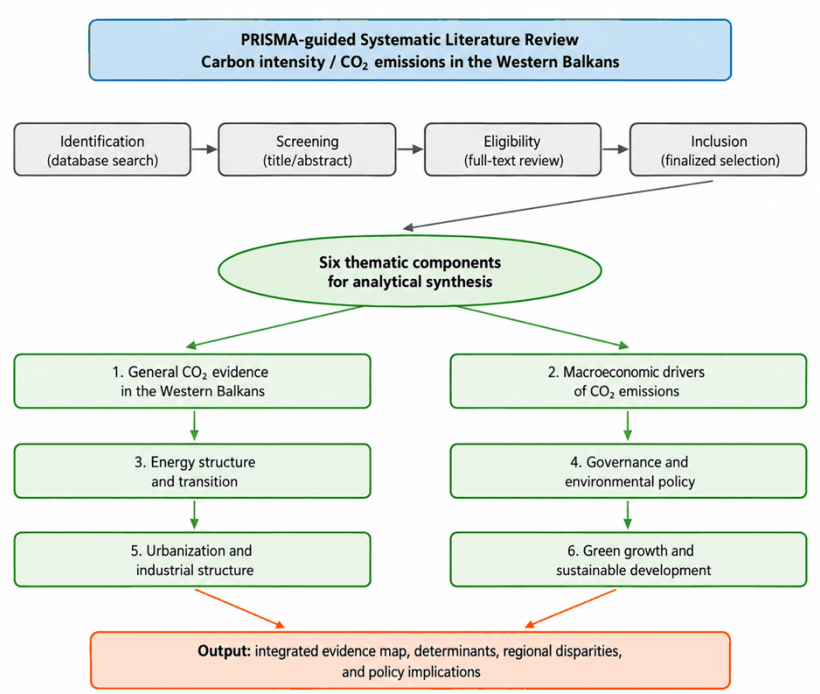


Figure 1. Conceptual framework—CO_2 emissions, carbon intensity, and sustainable economic growth literature in the Western Balkans

The Green Agenda for the Western Balkans, a regional initiative developed to complement the European Green Deal, underscores the Western Balkans' commitment to sustainable development (European Commission, 2019). The region's nations recognized the significance of shifting to a competitive, resource-efficient, and climate-neutral economy in November 2020 with Sofia Declaration, which formalized this commitment (Regional Cooperation Council, 2020b). In order to promote carbon neutrality and sustainable economic growth, the Western Balkans have committed to integrating environmental priorities into all sectors and policies through the implementation of the Green Agenda. Each pillar sets out specific objectives and strategies to support socioeconomic and environmental progress across the region. These commitments were further operationalized through adoption 2021–2030 Action Plan for the Green Agenda, which sets out seven roadmaps and 58 specific measures to support pollution control, carbon pricing, coal phase-out, and alignment with EU environmental policies, including the EU Emissions Trading System (ETS) (Regional Cooperation Council, 2021). The Western

Balkans aim to achieve several objectives through these initiatives, including reducing greenhouse gas emissions, promoting renewable energy use and circular economy practices, enhancing biodiversity conservation, and creating green jobs. Alignment with the European Green Deal help ensure that regional policies are consistent with broader international commitments, such as the Paris Climate Agreement and the Sustainable Development Goals (European Commission, 2019). The Green Agenda for the Western Balkans represents an important first step toward integrated sustainability into regional development plans. In line with EU regulations and international climate goals, the region is striving to strengthen both economic resilience and environmental stewardship by implementing the Sofia Declaration and the Action Plan. However, the implementation of the Green Agenda remains challenging due to limited institutional and regulatory capacity, insufficient funding, and limited understanding of the interconnected activities and guiding principles required to achieve the established objectives. Efficient transportation, infrastructure, and communication networks are essential for the economic development of the Western Balkans region and must be maintained and extended to meet evolving demands in a safe, cost-effective, and environmentally sustainable manner (Knez et al., 2022). Furthermore, Ziberi et al. (2025) used the EKC to analyze the implications of the European Green Agenda legal framework for the green economy in the case of the Western Balkan countries. The study applies the EKC hypothesis and uses data from the World Bank Indicators for the period 1990 to 2024. The findings indicate a U-shaped relationship between GDP per capita growth and CO_2 emissions, suggesting that higher GDP per capita growth is associated with higher CO_2 emissions. The study concludes that, the implications of the EKC hypothesis may vary across different stages and periods of a country's economic development and therefore may not be universally applicable. In addition, Mitić et al. (2022) demonstrated that CO_2 emissions are positively correlated with GDP, with each additional increase in GDP above the average leading to a 0.08% increase in CO_2 emissions. These findings indicate that over the past six decades, economic expansion has negatively affected the environment in the Western Balkans. Similarly, Mustafa et al. (2026) highlighted the importance of the green economy in the case of Kosovo, a Western Balkan country. The study investigates the role of green promotion in business success with particular emphasis on sustainability challenges, institutional support, and strategic priorities. Based on a survey of 397 businesses, the most powerful predictor of green promotion success is the perceived significance of the green transition. The findings suggest that organizations and policymakers can increase the effectiveness of green transition initiatives by providing targeted assistance to smaller businesses. Furthermore, it is worth noting that education plays an important role in promoting sustainable economic growth (Ziberi et al., 2022). In the new era of economic development, there is a need to update study programs (Vehapi & Ziberi, 2026) in order to equip the younger generations with the skills required to strengthen future economic resilience.

Vranjanac et al. (2018) examined the differences in per capita CO_2 emissions between Western Balkan countries and developed EU countries between 2006 and 2014. In contrast, Mesfun et al. (2018) investigated a power distribution system that uses low-carbon generation units, such as wind, solar, and biomass, across the EU and the Western Balkans. The findings indicate that, within a carbon price range of 0–200 €/tCO_2, renewable electricity can account for 9%–39% of the total generation mix. However, the optimal generation mix allows only a limited contribution from bio-electricity. Lower installed renewable energy capacity could increase reliance on natural gas as a result of the expansion of cross-border transmission capacity expansion. According to Ignjatović et al. (2024), Western Balkan countries should adopt a strategic approach to the green transition, emphasizing the restructuring and reform of the energy sector. They have improved energy efficiency but have not fully realized the potential of renewable energy. Because the circular economy generates less waste and improves resource productivity, it can accelerate the green transition. It is also suggested that cross-sectoral governance should be coordinated. Energy intensity continues to be a major source of environmental pressures, as evidenced by a recent study on Western Balkan countries that finds a negative correlation between GDP per capita and CO_2 emissions, a positive correlation with energy consumption, and no significant link with industrial production (Zhigolli & Fetai, 2024).

A thorough and reliable research methodology is required to address the cross-country heterogeneity and the multidimensional nature of the carbon intensity of GDP. A review of the existing literature reveals mixed and context-specific findings regarding the relationship between economic growth, energy use, and environmental outcomes. Thus, the primary research gap does not arise from a lack of evidence on the relationship between economic growth and environmental impacts in the Western Balkans, but rather from the absence of a systematic review that would enable the existing evidence to be interpreted on a comparable basis. The current body of literature remains fragmented with respect to definitions and indicators, the methodologies employed, and the units of analysis used to assess carbon intensity, absolute CO_2 emissions, environmental quality, and the carbon footprint. This inconsistency makes comparisons across studies and countries particularly challenging and makes it difficult to determine allow one to discern whether the differences found are truly country-specific or are attributable to methodological variation. This systematic review contributes to the literature by synthesizing the existing evidence through six different analytical perspectives and by identifying specific areas of consensus, divergence, methodological inconsistencies, and underrepresentation. Thus, while the review contributes to the existing knowledge base by synthesizing existing research findings, it also provides a new integrative analytical

framework for interpreting the decarbonization literature on the Western Balkans. In doing so, it lays the groundwork for future research, especially through the standardization of environmental indicators, the distinction between carbon intensity from absolute carbon emissions and the application of comparable country-level and sectoral models capable of explaining the heterogeneous environmental effects of economic growth across the region. The following section presents the research methodology, including the search strategy, the PRISMA-guided review framework, and the eligibility criteria applied to the inclusion and exclusion of studies.

2. Research Methodology

This study is based on a PRISMA-guided systematic literature review that aims to synthesize empirical evidence on the factors that influence CO_2 emissions, carbon intensity and sustainable economic growth in the Western Balkans as well as on regional disparities within the broader context of sustainable economic development across the region. This systematic review approach was chosen to ensure transparency and replicability through a systematic database search, predefined inclusion criteria, and structured data extraction. In accordance with the PRISMA guidelines (PRISMA, 2020), the studies were screened for inclusion into the review based on their alignment with the research questions. This synthesises the existing literature on the field of carbon intensity, CO_2 emissions, macroeconomic factors, energy balance and transition, governance and environmental policy, urbanization and industrial structure, green transition, renewable energy, decarbonization, and sustainable development. In order to supplement the qualitative synthesis, a network visualization was created through the VOSviewer software. The data used to create this visualization were obtained through the systematic coding of the 30 studies included in this systematic review. Each source was coded according to its area of interest, the key themes influencing the outcomes and the outcomes itself. The 30 studies retained after the PRISMA screening process are presented in Appendix A, Table A. The information extracted from these studies also served as the coding basis for the construction of the VOSviewer network. The following subsection describes the literature search strategy including the Scopus search strings and the strategies and the inclusion criteria used in this systematic review.

2.1 Data Extraction Process and Thematic Coding Procedure

A search strategy was implemented in the Scopus database to identify relevant literature on CO_2 emissions, carbon intensity, and sustainable economic development in the Western Balkans. Scopus was chosen because it was the only multidisciplinary database available to the authors, whereas the Web of Science database, which could not be accessed at that time. Therefore, the use of a single database is acknowledged as a limitation of the study. In addition, relying solely on the Scopus database may have affected the inclusion of relevant articles related to CO_2 emissions, carbon intensity, and sustainable economic growth in the Western Balkans. This is particularly important because the scope of the review was geographically limited to the region. This means that there may be studies focusing on individual Western Balkan countries that are published in regional journals not indexed in Scopus. Hence, the selected number of studies may not capture the full body of literature on this subject. Future systematic reviews could incorporate additional bibliographic databases, such as Web of Science and Google Scholar to enhance literature coverage and reduce the potential risk of database-related selection bias.

The process of identification, screening, eligibility, and selection of articles was concluded in accordance with PRISMA 2020 guidelines, as shown in the PRISMA flow diagram. For transparency and reproducibility, the full search strings, Boolean terms, and thematic keywords utilized for all six themes are provided in Table 1. The data from the 30 selected articles were then manually extracted into an Excel spreadsheet containing the following elements: author and year, country or region, study design, period, outcome examined, main variables, main research focus, key findings, main contribution, and DOI. During the full-text screening process, the 30 studies were grouped into six themes based on their main research focus and the key ideas found in the literature. Articles covering multiple aspects were allocated to the theme they primarily addressed, while all the relevant results were considered during the synthesis. The extracted data were also used as the basis for constructing the two VOSviewer maps presented in the paper.

Table 1 presents the key Scopus search strings developed in accordance with the PRISMA guidelines for this systematic review. The literature search strategy of this study was guided by six research questions. First, the review sought to identify relevant evidence on general issues related to CO_2 emissions and carbon intensity in the Western Balkan countries. The study also examined literature focusing on the macroeconomic factors influencing CO_2 emissions, carbon intensity and sustainable economic growth. including economic growth, GDP per capita, financial development, inflation, FDI, and trade openness. Furthermore, the review examined issues related to energy structure and energy transition, including renewable energy, energy consumption, electricity consumption, fossil fuels, energy efficiency, and decarbonization. In addition, the study reviewed literature on governance and environmental policy, including institutional quality, environmental policy, carbon taxation, carbon pricing, the EU Green Agenda, and CBAM. The review also sought to identify studies related to urbanization and industrial

structure, including industrialization, construction, manufacturing, and industrial value added, as well as studies addressing green growth and sustainable development.

Table 1. Scopus search string strategies

No.	Thematic Themes	Scopus Search String
1	General CO_2 evidence in the Western Balkans	TITLE-ABS-KEY ((“CO_2” OR “carbon dioxide” OR “carbon emissions” OR “CO_2 emissions”) AND (“Western Balkans” OR Albania OR Kosovo OR Serbia OR Montenegro OR “North Macedonia” OR “Bosnia and Herzegovina”))
2	Macroeconomic drivers of CO_2 emissions	TITLE-ABS-KEY ((“CO_2 emissions” OR “carbon intensity” OR “carbon emissions”) AND (“economic growth” OR GDP OR “GDP per capita” OR “financial development” OR inflation OR FDI OR “trade openness”) AND (“Western Balkans” OR Albania OR Kosovo OR Serbia OR Montenegro OR “North Macedonia” OR “Bosnia and Herzegovina”))
3	Energy structure and transition	TITLE-ABS-KEY ((“CO_2 emissions” OR “carbon intensity” OR “carbon emissions”) AND (“renewable energy” OR “energy consumption” OR “electricity consumption” OR “fossil fuel” OR “energy efficiency” OR decarbonization) AND (“Western Balkans” OR Albania OR Kosovo OR Serbia OR Montenegro OR “North Macedonia” OR “Bosnia and Herzegovina”))
4	Governance and environmental policy	TITLE-ABS-KEY ((“CO_2 emissions” OR “carbon intensity” OR “carbon emissions”) AND (“institutional quality” OR governance OR “environmental regulation” OR “carbon tax” OR “carbon pricing” OR “EU Green Agenda” OR CBAM) AND (“Western Balkans” OR Albania OR Kosovo OR Serbia OR Montenegro OR “North Macedonia” OR “Bosnia and Herzegovina”))
5	Urbanization and industrial structure	TITLE-ABS-KEY ((“CO_2 emissions” OR “carbon intensity” OR “carbon emissions”) AND (urbanization OR industrialization OR “industrial structure” OR construction OR manufacturing OR “value added”) AND (“Western Balkans” OR Albania OR Kosovo OR Serbia OR Montenegro OR “North Macedonia” OR “Bosnia and Herzegovina”))
6	Green growth and sustainable development	TITLE-ABS-KEY ((“green growth” OR “sustainable development” OR “sustainable economic growth” OR “green transition”) AND (“CO_2 emissions” OR “carbon intensity” OR “carbon emissions” OR decarbonization) AND (“Western Balkans” OR Albania OR Kosovo OR Serbia OR Montenegro OR “North Macedonia” OR “Bosnia and Herzegovina”))

Table 2 presents the PRISMA-informed search strategy employed in conducting the systematic literature review. The literature search was conducted using the Scopus database following the PRISMA 2020 guidelines to enhance transparency throughout the screening and selection process. The search strategy focused on relevant studies addressing carbon intensity, CO_2 emissions intensity, sustainable development, urbanization, environmental policy frameworks, and green growth. Journal articles, conference papers, and book chapters retrieved from Scopus were considered for inclusion based on the predefined eligibility criteria. Conference papers and book chapters were included only if they explicitly addressed the Western Balkan region or provided relevant regional data and information regarding the research objectives, methodologies, variables, environmental impacts, and key findings. Studies were excluded if they lacked sufficient methodological or empirical details, were unrelated to the research question, or did not provide relevant evidence on CO_2 emissions, carbon intensity, decarbonization and sustainable economic growth.

Table 2. PRISMA search strategy components

PRISMA Search Strategy Component	Description
Database	Scopus
Search approach	Broad systematic screening search following the PRISMA Statement framework
Purpose of the search string	To identify a broad body of literature related to carbon intensity, CO_2 emissions intensity, sustainable development, and green economic growth for the initial PRISMA identification and screening stages
Screening focus	Studies examining the relationship between carbon intensity, CO_2 emissions intensity, sustainable development, urbanization, policy frameworks and green growth
Initial search outcome	The search generated 790 records before the screening and eligibility assessment process
Main inclusion focus	Peer-reviewed studies related to carbon intensity, sustainable growth, green transition, and environmental-economic sustainability in the case of Western Balkan countries
Main exclusion focus	Not directly related to CO_2 emissions, carbon intensity, carbon emissions, or decarbonization; Insufficient empirical or methodological information; Not aligned with the six thematic dimensions of the review; Conceptual, descriptive, or policy papers without sufficient analytical relevance; Overlapping or replaced versions of already included studies; Methodological limitations

Note: PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

The literature search was conducted using search for the literature was carried out via the Scopus database in July 2026 following a systematic literature review approach in accordance with PRISMA guidelines. The search process was based on six related Boolean search strings that covered different thematic aspects of the review, namely, general evidence on CO_2 in the Western Balkans, macroeconomic determinants of CO_2 emissions, carbon intensity, energy transition and transformation, governance and environmental policies, urbanization and industrial structure, and green growth and sustainable development. In the identification stage, the first search string, TITLE-ABS-KEY ((“CO_2” OR “carbon dioxide” OR “carbon emissions” OR “CO_2 emissions”) AND (“Western Balkans” OR Albania OR Kosovo OR Serbia OR Montenegro OR “North Macedonia” OR “Bosnia and Herzegovina”)), generated 486 documents in Scopus. This broad search was designed to capture general evidence on CO_2-related research in the Western Balkan region. The second search string, TITLE-ABS-KEY ((“CO_2 emissions” OR “carbon intensity” OR “carbon emissions”) AND (“economic growth” OR GDP OR “GDP per capita” OR “financial development” OR inflation OR FDI OR “trade openness”) AND (“Western Balkans” OR Albania OR Kosovo OR Serbia OR Montenegro OR “North Macedonia” OR “Bosnia and Herzegovina”)), generated 35 documents and focused on macroeconomic drivers of CO_2 emissions. The third thematic search, focusing on energy structure and transition, yielded 149 documents, while the fourth search, focusing on governance and environmental policy, yielded 17 documents.

The fifth search string, TITLE-ABS-KEY ((“CO_2 emissions” OR “carbon intensity” OR “carbon emissions”) AND (urbanization OR industrialization OR “industrial structure” OR construction OR manufacturing OR “value added”) AND (“Western Balkans” OR Albania OR Kosovo OR Serbia OR Montenegro OR “North Macedonia” OR “Bosnia and Herzegovina”)), generated 37 documents and focused on urbanization and industrial structure. Finally, the sixth search string, TITLE-ABS-KEY ((“green growth” OR “sustainable development” OR “sustainable economic growth” OR “green transition”) AND (“CO_2 emissions” OR “carbon intensity” OR “carbon emissions” OR decarbonization) AND (“Western Balkans” OR Albania OR Kosovo OR Serbia OR Montenegro OR “North Macedonia” OR “Bosnia and Herzegovina”)), generated 66 documents related to green growth, sustainable development, green transition, decarbonization, and carbon-related outcomes. In total, the six Scopus searches yielded 790 records before duplicate removal and screening. Due to the expected overlap between different search strings, the initial result represents the gross number of records retrieved and does not reflect the number of unique studies included in the review. The records were screened according to predefined eligibility criteria. Only studies that addressed the Western Balkans or at least one of the selected countries, examined CO_2 emissions, carbon intensity, carbon emissions, decarbonization or other carbon-related environmental outcomes, and provided information relevant to at least one of the six review themes were considered eligible. Studies that fell outside the scope of this systematic review or lacked sufficient scientific or empirical information were excluded from further consideration.

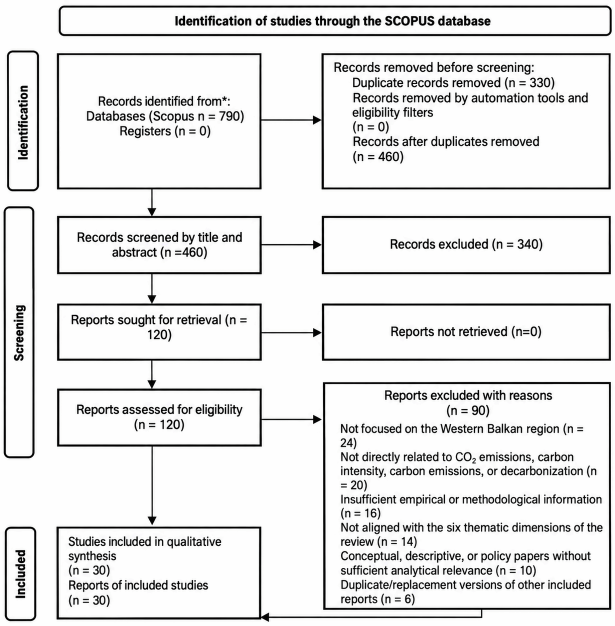


Figure 2. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram of the literature selection process on carbon intensity and sustainable economic growth

Figure 2 illustrates the PRISMA flow diagram, outlining the systematic process of identifying, screening,

assessing eligibility, and selecting studies on carbon intensity and sustainable economic growth. The literature search was conducted using Scopus database and six Boolean search strings developed in accordance with the thematic scope of the review. The literature search yielded 790 records, while no records were identified through registers (n = 0). Due to the similarities and partial overlaps between the search strings, duplicate records were identified among the search results. Before the screening process, 330 duplicate records were removed. No records were removed using automation tools or for other reasons before screening (n = 0). Thus, 460 records remained for title and abstract screening.

During this stage, 340 records were excluded because they lacked relevant information regarding the regional focus, carbon-related outcomes, or the thematic dimensions of the review. Following title and abstract screening, 120 reports were sought for retrieval. All reports were successfully retrieved, with no reports remaining unretrieved (n = 0). Consequently, 120 full-text reports were assessed for eligibility. Of these 120 reports, 90 reports were excluded for the following reasons: 24 reports did not specifically address the Western Balkans region; 20 reports did not focus on CO_2 emissions, carbon intensity, carbon emissions, or decarbonization; 16 reports lacked sufficient empirical or methodological information; 14 reports were not aligned with the six thematic dimensions of the review, 10 reports were conceptual, descriptive, or policy papers that lacked analytical relevance; and 6 reports were duplicate or alternative versions of other included reports. The study selection process, conducted in accordance with the PRISMA 2020 guidelines, resulted in the inclusion of 30 studies in the final qualitative synthesis. These studies provided the basis for synthesizing evidence on the determinants of carbon intensity, the environmental effects of CO_2 emissions, regional disparities, and sustainable economic development in the Western Balkans. The following section presents the findings derived from the systematic synthesis of the 30 studies included in the analysis, as reported in accordance with the PRISMA 2020 guidelines.

3. Results

The findings of the systematic review were derived from 30 studies that met the predefined eligibility criteria and were selected through the PRISMA-guided screening process. The selected studies vary in geographical scope, research methodology, period of analysis, environmental measures, explanatory factors and analytical methods employed. To present the findings in a structured manner, the evidence was categorized into six main themes corresponding to the key economic, energy, institutional, financial, sectoral and sustainability dimensions identified in the literature. It is important to note that the findings are distinguished according to whether the studies examine absolute CO_2 emissions, CO_2 emissions per capita, carbon intensity or other related environmental outcomes.

Table 3 presents an overview of studies examining the relationship between CO_2 emissions, carbon intensity, economic growth and sustainable development. The reviewed literature indicates that economic growth alongside energy consumption and urbanization, influences the long-term dynamics of carbon emissions in the region. These factors also represent key determinants of environmental degradation in the Western Balkans.

Table 3. Theme 1—general CO_2 emissions, carbon intensity, and sustainable growth

No.	Author & Year	Country/ Region	Study Design/Method	Period	Outcome Examined	Main Variables
1	Ziberi et al. (2025)	Six Western Balkan countries: Albania, Bosnia and Herzegovina, Kosovo, Montenegro, North Macedonia, and Serbia	Quantitative panel-data analysis based on the Environmental Kuznets Curve (EKC) framework	1990–2024	CO_2 emissions	Carbon intensity, GDP per capita growth, and squared GDP per capita growth
2	Mitić et al. (2024)	Serbia	autoregressive distributed lag (ARDL) modelling of the EKC and bootstrap logistic regression	1995–2019	CO_2 emissions and environmental quality	Economic growth, electricity consumption, and trade openness
3	Obradović et al. (2025)	Western Balkans	Panel cointegration analysis and	2001–2022	CO_2 emissions and environmental quality	Economic growth, energy consumption, and urbanization

4	Oprea & Bâra (2026)	Serbia	pooled mean group ARDL (PMG-ARDL) Time-series analysis using a vector error correction model (VECM)	From 1995	Economic growth, CO_2 emissions, and renewable-energy consumption	GDP per capita, CO_2 emissions, renewable-energy consumption, urbanization, and trade openness
5	Methmini et al. (2025)	Twenty high-emitting countries across five continents, including Albania	Comparative panel-data study using multiple linear regression	2000–2021	CO_2 emissions	GDP, energy consumption, and trade openness

No.	Author & Year	Main Focus	Key Findings	Main Contribution	DOI
1	Ziberi et al. (2025)	The relationship between economic growth and carbon intensity within the context of the EU Green Agenda legal framework	The study identifies a U-shaped relationship between GDP per capita growth and carbon intensity. Above the estimated turning point of 4,259.69, higher GDP per capita growth is associated with increased carbon intensity, indicating that the conventional inverted-U EKC hypothesis is not supported for the Western Balkan sample	Provides regional panel evidence that the EKC relationship is not universal and highlights the need to align economic growth with the EU Green Agenda and stronger environmental policies in the Western Balkans	https://doi.org/10.3846/btp.2025.23716
2	Mitić et al. (2024)	The long-run and short-run relationships between economic growth, electricity consumption, trade openness, and CO_2 emissions in Serbia	The long-run results confirm an inverted U-shaped relationship between economic growth and CO_2 emissions, supporting the conventional EKC hypothesis. Economic growth, electricity consumption, and trade openness have statistically significant long-run effects on CO_2 emissions. In the short run, a causal relationship is identified only	Provides combined long-run, short-run, and predictive evidence on the determinants of CO_2 emissions in Serbia, highlighting the importance of renewable-energy investment, sustainable trade policies, and long-term emission-reduction measures	https://doi.org/10.1007/s11356-023-31617-y

			from electricity consumption to CO_2 emissions. The bootstrap logistic regression also indicates that all three factors significantly influence environmental quality		
3	Obradović et al. (2025)	The long-run and short-run effects of economic growth, energy consumption, and urbanization on carbon emissions in the Western Balkans	Economic growth, energy consumption, and urbanization have statistically significant long-run effects on environmental quality. In the short run, only economic growth has a statistically significant effect. The results indicate that energy consumption, urbanization, and economic expansion contribute to long-term environmental degradation	Provides regional evidence that accelerated urbanization, increasing energy consumption, and economic growth jointly shape long-term CO_2 emissions in the Western Balkans, highlighting the need for improved energy efficiency, sustainable urban planning, and pollution-mitigation policies	https://doi.org/10.3390/urbansci9040119
4	Oprea & Bâra (2026)	Short-run and long-run relationships between economic growth and environmental sustainability in Serbia	The results indicate a short-run trade-off between economic growth and environmental sustainability, as lagged CO_2 emissions and renewable-energy consumption negatively affect GDP growth. In the long run, GDP growth is positively associated with trade openness and urbanization, while renewable-energy consumption is strongly influenced by urbanization. CO_2 emissions evolve relatively independently of trade openness	Demonstrates that Serbia's growth–sustainability relationship differs between the short and long run and identifies urbanization as an important mechanism linking economic growth, renewable-energy integration, and system adjustment	https://doi.org/10.3846/jbem.2026.26926

			and urbanization. GDP, CO_2 emissions, and renewable-energy consumption significantly adjust to deviations from long-run equilibrium Energy consumption significantly increases carbon emissions in almost all examined countries, with Panama and Kazakhstan reported as exceptions. Albania is identified among the European countries showing a statistically significant contribution to regional carbon-emission patterns		
5	Methmini et al. (2025)	The effects of economic growth, energy consumption, and trade openness on carbon emissions across major emitting countries		Provides comparative international evidence that energy consumption is a major driver of carbon emissions and supports the adoption of country-specific strategies that reconcile economic development, energy use, trade, and environmental objectives	https://doi.org/10.1080/17565529.2024.2375737

The relationship between economic growth and environmental degradation in the Western Balkans is nonlinear, context-dependent, and sensitive to the environmental indicators and methodologies. Ziberi et al. (2025), using panel data from six Western Balkan countries covering the period 1990–2024, identified a U-shaped relationship between GDP per capita growth and CO_2 emissions. Based on their findings, economic growth beyond the estimated threshold of 4,259.69 is associated with an increase in carbon intensity. In contrast, Mitić et al. (2024) provided empirical evidence supporting the traditional inverted-U EKC hypothesis for Serbia over the period 1995–2019. According to their long-run estimates, economic growth, electricity consumption, and trade openness significantly influence CO_2 emissions, whereas the relationship between electricity consumption and emissions is also supported by short-run causality tests. Further supporting evidence at the regional level is provided by Obradović et al. (2025), who emphasized the importance of energy consumption and urbanization as long-term determinants of environmental degradation. Their study, employing panel cointegration and pooled mean group autoregressive distributed lag (PMG-ARDL) methods for the Western Balkans over the period 2001–2022, demonstrates that economic growth, energy consumption, and urbanization significantly affect environmental quality in the long run, whereas only economic growth has a significant effect in the short run. These findings suggest that the environmental impacts of urbanization and energy consumption may emerge gradually over time.

Additional evidence of temporal heterogeneity in Serbia is provided by Oprea & Bâra (2026). Using a vector error correction model (VECM), they identify a short-run trade-off between economic growth and environmental sustainability, while long-run relationships are shaped by factors such as trade openness, urbanization, and renewable energy consumption. Their finding that CO_2 emissions are independent of trade openness and urbanization in some model specifications contrasts with long-run relationships reported by Obradović et al. (2025). This discrepancy may be partly explained by differences in methodological approaches, particularly the use of a single-country time-series analysis rather than a regional panel-data approach. The broader comparative findings of Methmini et al. (2025), whose study included Albania alongside 20 other major emitting countries, further support the role of energy consumption as one of the primary determinants of carbon emissions.

Energy consumption emerges as the factor most consistently associated with higher levels of CO_2 emissions and environmental degradation. The impact of economic growth is not universal; the relationship between economic growth and environmental sustainability may follow an inverted-U-shaped or U-shaped pattern or exhibit dynamic characteristics, depending on the country examined, the time period considered, the environmental outcome variable, and the analytical method employed. In addition, the effects of urbanization and

trade openness may operate through indirect and longer-term mechanisms, with their environmental impacts depending on energy efficiency. Consequently, the principal research gap in this context is not the lack of evidence on the relationship between economic growth and the environment, but rather the limited availability of comparable evidence based on harmonized concepts and indicators. Existing studies adopt different definitions and measurements of CO_2 emissions, environmental quality, and carbon footprint, making cross-study comparisons challenging. Future research should prioritize the harmonization of environmental indicators, distinguish carbon intensity from absolute emissions, and apply comparable country-level or sectoral models to explain differences in the environmental consequences of economic growth.

Theme 2, presented in Table 4, highlights that energy structure is one of the most important factors influencing CO_2 emissions and carbon intensity in the Western Balkans. The reviewed literature provides substantial dependence on fossil fuels remains high in the Western Balkans due to the region's reliance on coal-based electricity generation. However, the findings also indicate increasing opportunities for investments in renewable energy, solar photovoltaic systems, solar thermal power, hydropower, and energy efficiency improvements. The reviewed studies focusing on Bosnia and Herzegovina, Kosovo, North Macedonia, and Albania, suggest that renewable energy use contributes to emissions reductions and decarbonization efforts. Furthermore, the reviewed literature indicates that although technical potential exists, energy transition also depends on investment capacity and long-term planning.

The studies reviewed in this paper generally identify renewable energy use, energy efficiency, and technological advancement as key factors driving the decarbonization process in the Western Balkans. Nevertheless, the evidence indicates that the feasibility, financial implications, and environmental impacts of the energy transition vary considerably across countries, sectors, and technologies. These differences reflect variations in the existing energy mix, dependence on coal and hydropower, infrastructure, investment availability, regulatory readiness, and renewable energy potential.

Table 4. Theme 2—energy transition, renewable energy, and decarbonization

No.	Author & Year	Country/ Region	Study Design/Method	Period	Outcome Examined	Main Variables
6	Ćosić & Duić (2025)	Croatia and Bosnia and Herzegovina	Energy-system modelling using the Dispa-SET unit-commitment and power-dispatch model	2025–2035; National Energy and Climate Plans (NECP), low-renewable integration (LRES), and high-renewable integration (HRES) scenarios	CO_2 intensity, fossil-fuel dependence, electricity trade, renewable-energy curtailment, and system flexibility	carbon border adjustment mechanism (CBAM) compliance, variable renewable-energy deployment, coal-power operation, electricity exports, and renewable-energy curtailment
7	Zakić et al. (2026)	Serbia, with comparative reference to China and the European Union (EU)	Book chapter; qualitative policy and investment-readiness analysis supported by a historical comparative approach	Historical and contemporary policy analysis	Renewable-energy transition readiness and economic-growth potential	Renewable-energy investment, regulatory framework, fossil-fuel dependence, energy and carbon intensity, external financing, EU integration, and Sino-Serbian energy cooperation
8	Hasović et al. (2015)	Bosnia and Herzegovina	Energy-system modelling using the EnergyPLAN model	Scenarios projected up to 2020	CO_2 emissions, renewable-energy target compliance, electricity production, and energy-system performance	New thermal-power investments, hydrological conditions, hydropower development, retirement of thermal units, electricity demand, and renewable-energy deployment
9	Mustafa et al. (2026)	North Macedonia	Time-series analysis using a vector autoregressive model	2000–2023	CO_2 emissions, economic growth, and inflation	Renewable-energy investments/capacity, CO_2 emissions, GDP, inflation, and regulatory index
10	Gjukaj et al. (2024)	Kosovo, with a case study of the Sibovc coal basin	Renewable-energy feasibility and environmental-impact assessment using PVsyst and	Scenario-based analysis; study period not specified	Solar-energy generation potential, projected CO_2 emission reduction, and environmental feasibility	Solar PV deployment, degraded-land reuse, available land area, solar-energy potential, electricity demand, and projected environmental benefits

11	Špago et al. (2026)	Bosnia and Herzegovina; five cities from central, northern, and southern regions	RETScreen simulations Techno-economic and environmental assessment based on long-term solar-irradiance data	Long-term solar-resource assessment; specific data period not reported in the abstract	CO_2 emission reduction, solar-energy coverage, household energy-cost savings, and investment payback period	Solar irradiance, geographical location, household hot-water demand, solar thermal collectors, displaced energy source, investment costs, and energy prices
12	Belussi et al. (2021)	Albania; public-school case study in Tirana	Building energy-performance assessment and retrofit scenario analysis	Existing-building assessment and proposed retrofit scenarios; specific data period not reported	Primary energy performance and feasibility of achieving the nearly Zero Energy Building standard	Building thermal performance, retrofit technologies, electricity-mix characteristics, Primary Energy Factors, hydropower-based electricity, and national energy-market conditions

No.	Author & Year	Main Focus	Key Findings	Main Contribution	DOI
6	Ćosić & Duić (2025)	The combined effects of CBAM and variable renewable-energy integration on interconnected fossil-based electricity systems	Accelerated renewable-energy deployment reduces fossil-fuel dependence and CO_2 intensity. Under the HRES scenario, the reported CO_2 intensity in Bosnia and Herzegovina decreases to 0.85 gCO_2/kWh by 2035. Electricity exports to Croatia decline under the NECP scenario, while high renewable integration stabilises exports. However, renewable-energy curtailment reaches 2.73 TWh per year, equivalent to 24.2% of variable renewable energy sources (VRES) generation, indicating substantial storage and grid-flexibility constraints	Demonstrates that renewable-energy integration and carbon pricing can accelerate regional decarbonization, but also create challenges related to curtailment, grid flexibility, storage, electricity trade, and energy security	https://doi.org/10.1016/j.ecmx.2025.101008
7	Zakić et al. (2026)	The potential of renewable-energy cooperation with China to support Serbia's energy transition and economic growth	Serbia's dependence on fossil fuels and its high energy and carbon intensity create substantial transition challenges. Limited access to EU cohesion	Connects renewable-energy cooperation, investment policy, and EU accession requirements, proposing Sino-Serbian cooperation as a potential mechanism for accelerating Serbia's energy	https://doi.org/10.1007/978-3-032-04071-8_16

			funds increases the importance of external investment. Cooperation with China may support renewable-energy deployment, technology transfer, and investment, provided that such cooperation is aligned with Serbia's EU energy and environmental obligations	transition and supporting sustainable economic growth	
8	Hasović et al. (2015)	The impact of planned power-plant investments and hydrological variability on Bosnia and Herzegovina's energy system and renewable-energy commitments	Delays in planned power-sector investments and variations in hydrological conditions substantially affect electricity production and overall system performance. Continued investment in coal-based thermal plants increases dependence on fossil fuels and creates difficulties in meeting renewable-energy targets. Bosnia and Herzegovina need to diversify beyond large hydropower and expand other renewable-energy sources	Demonstrates through scenario modelling that fossil-fuel-oriented investment and hydrological dependence may constrain decarbonization, while broader renewable-energy diversification is necessary to improve system resilience and meet international commitments	https://doi.org/10.2298/TSCI150105042H
9	Mustafa et al. (2026)	The impact of renewable-energy investment on the decarbonization process of North Macedonia	Renewable-energy capacity expansion has a negative and statistically significant relationship with CO_2 emissions, confirming its contribution to decarbonization. However, the causality tests do not provide evidence that renewable-energy investments cause changes in	Provides country-specific empirical evidence that investment in solar, wind, and hydropower supports emissions reduction in North Macedonia, while showing that its short-term macroeconomic effects are not clearly established	https://doi.org/10.1007/978-3-032-04214-9_18

			economic growth or inflation		
10	Gjukaj et al. (2024)	The potential use of mining-degraded land for solar photovoltaic development and Kosovo's energy transition	The simulation results indicate that solar PV parks installed on degraded mining land could generate renewable electricity and substantially reduce CO_2 emissions. The proposed pilot project on an 80-hectare parcel in the Sibovc coal basin also demonstrates the potential for future expansion	Demonstrates how degraded coal-mining land can be repurposed for renewable-energy infrastructure, simultaneously supporting decarbonization, land rehabilitation, pollution reduction, and sustainable energy development in Kosovo	https://doi.org/10.12912/27197050/189603
11	Špago et al. (2026)	The technical, environmental, and economic feasibility of residential solar thermal collectors for domestic hot-water production	Solar thermal systems can cover 30%–60% of annual household hot-water demand in central and northern Bosnia and Herzegovina and more than 80% in southern areas, reaching full coverage during summer. Replacing electricity, coal, and natural gas produces substantial CO_2 emission reductions, particularly when coal- and electricity-based heating are displaced. Annual household savings range from BAM 200 to BAM 500, with estimated payback periods of 6–10 years	Provides location-specific evidence that residential solar thermal systems can simultaneously reduce household CO_2 emissions and energy costs, while demonstrating that their feasibility varies according to solar conditions and the conventional energy source being replaced	https://doi.org/10.1007/978-3-032-30107-9_13
12	Belussi et al. (2021)	The potential to retrofit an existing public-school building in Tirana to achieve the nearly Zero Energy Building standard	Albania's hydropower-dominated electricity system provides comparatively low-carbon electricity and creates favorable conditions for nearly Zero-Energy Buildings. The study	Provides a sector-specific pathway for building decarbonization in Albania by linking retrofit strategies, national electricity characteristics, and EU energy-efficiency requirements. It also develops locally relevant Primary Energy Factors	https://doi.org/10.1051/e3sconf/202131202005

	indicates that appropriate retrofit technologies and energy-efficiency measures can substantially improve the school's energy performance and support achievement of the nearly Zero-Energy Building target	through comparison with neighboring countries	

At the regional electricity system level, Ćosić & Duić (2025) demonstrated through their modelling results that rapid integration of variable renewable energy sources could significantly reduce dependence on fossil fuels and the CO_2 emissions associated with electricity generation in the interconnected electricity systems of Croatia and Bosnia and Herzegovina. Under the high-renewables scenario, CO_2 emissions from electricity generation in Bosnia and Herzegovina are projected to decline substantially by 2035. However, the authors also identify significant challenges, as renewable energy curtailment is projected to reach 24.2% in 2035. These findings suggest that the environmental benefits of rapid renewable energy deployment may be constrained by the electricity system's limited capacity to integrate variable renewable generation.

The findings of Hasović et al. (2015) further highlighted the importance of structural characteristics and energy investments. Based on scenarios developed using the EnergyPLAN model, Bosnia and Herzegovina may face challenges in meeting its renewable energy commitments, given its continued investment in coal-fired power generation and substantial dependence on hydrological conditions. Despite its considerable hydropower and biomass potential, the country's reliance on large-scale hydropower plants creates vulnerability to variations in hydrological conditions between wet and dry periods. Taken together, the findings of Ćosić & Duić (2025) and Hasović et al. (2015) suggested that Bosnia and Herzegovina must address two interrelated challenges: decarbonization and diversification of its energy mix.

Country-specific evidence from North Macedonia provides further support for the relationship between renewable energy investment and emissions reductions. According to Mustafa et al. (2026), renewable energy capacity development is negatively associated with CO_2 emissions, suggesting that investments in renewable energy contribute to the country's decarbonization efforts. This is an important finding, as it highlights the environmental benefits of renewable energy investments while leaving their broader macroeconomic implications less clearly established. This does not imply that such investments lack economic significance; rather, it indicates that their effects on employment, productivity, industrial development, energy dependence, and overall economic performance require further investigation using more detailed and comprehensive analytical models.

This empirical evidence offers a different perspective from the policy-oriented analysis of Zakić et al. (2026), which presents Serbia–China cooperation in renewable energy as a potential driver of economic development and innovation. The authors argue that Serbia's dependence on fossil fuels, high energy and carbon intensity, and limited access to EU cohesion funds increase the importance of foreign investment. In this context, cooperation with China may support renewable energy deployment and technology transfer, provided that such investments remain consistent with Serbia's EU accession commitments. The differences between the findings of Mustafa et al. (2026) and Zakić et al. (2026) may therefore be explained, at least partly, by differences in research design and analytical perspective.

At the project and technology level, Gjukaj et al. (2024) demonstrated that land degraded by coal-mining activities in Kosovo can be repurposed for solar photovoltaic electricity generation, while Špago et al. (2026) provided further evidence of the technical and economic feasibility of decentralized solar energy applications in Bosnia and Herzegovina. Solar thermal systems for households are estimated to meet 30%–60% of domestic hot water demand in the central and northern parts of the country and more than 80% in the south. The greatest environmental benefits are expected when solar thermal energy replaces coal-based or electricity-based water heating, whereas estimated household cost savings and payback periods indicate economic feasibility. At the same time, substantial geographical variation in energy availability and system performance suggests that the efficiency and economic benefits of renewable energy technologies depend on location-specific conditions.

Belussi et al. (2021) provided further evidence from the construction sector by assessing the feasibility of improving the energy performance of a public-school building in Albania to meet nearly Zero-Energy Building standards. Albania's favorable electricity mix, largely based on hydropower, which is a relatively low-carbon energy source, supports the transition towards more energy-efficient building operations. Thus, the study demonstrates the importance of suitable retrofitting technologies, improved thermal performance, and accurate

calculations of primary energy factors in meeting European energy efficiency standards. Unlike the previously reviewed studies focusing on electricity generation, this study emphasizes the role of reducing energy demand in achieving decarbonization. The literature consistently indicates the environmental benefits of renewable energy deployment and improvements in energy efficiency. These benefits include lower greenhouse gas emissions, reduced dependence on fossil fuels, and improved energy performance. Nevertheless, the reviewed studies also emphasize that renewable energy potential alone does not guarantee successful decarbonization, as grid congestion, energy curtailment, hydrological variability, continued investment in fossil fuels, financial and regulatory constraints, geographical conditions, and technological availability may influence the extent of environmental benefits.

The principal research gap is the limited integration of technological, economic, institutional, and broader socioeconomic dimensions of the energy transition within a unified analytical framework. Most studies focus on individual technologies, specific countries, or particular scenarios. However, few studies systematically compare the costs, implementation barriers, and emissions reductions associated with renewable energy deployment across all Western Balkan countries. Moreover, the existing literature primarily comprises modelling studies, feasibility assessments, case studies, and policy analyses, with limited empirical evidence from completed renewable energy projects. Therefore, future research should evaluate the actual operational performance of these projects, including energy storage requirements and the electricity grid flexibility necessary for their effective integration.

Table 5 presents Theme 3, which examines the role of governance and environmental policies in shaping decarbonization outcomes. The reviewed studies exhibit considerable methodological diversity, employing panel data models, time-series analysis, dynamic estimation methods, linear programming models, energy systems modelling, risk index analyses, techno-economic analyses, and policy analyses. It is important to note that the reviewed studies ado not focus on a single type of environmental outcome. While some studies directly examine carbon intensity, others focus absolute or per capita CO_2 emissions, emissions from energy systems decarbonization performance, environmental quality, transition risks, abatement costs, or the economic impacts of climate policies.

Table 5. Theme 3—environmental governance, institutions, and climate policy

No.	Author & Year	Country/ Region	Study Design/Method	Period	Outcome Examined	Main Variables
13	Eicke et al. (2021)	Global and Bosnia and Herzegovina case	Mixed-method study: quantitative risk-index analysis and qualitative case studies	–	Carbon border adjustment mechanism (CBAM) within the European Green Deal	Export structure, emissions intensity, emissions-reduction targets, trade diversification, carbon lock-in, and monitoring, reporting, and verification (MRV) capacity
14	Lobanov & Lobanova (2026)	Western Balkans, with detailed estimates for Serbia	policy-impact analysis, CBAM	–	Assessing Impacts on Production and Exports of 35 Carbon-Intensive Goods. CBAM-related production, export exposure, and compliance costs	Export volumes, EU market dependence, export significance, trade relationships, production capacity, carbon pricing, embedded emissions, direct emissions, projected CBAM certificate prices
15	Krasniqi & Gara (2025)	Western Balkans	Panel-data econometric study	2010–2023	CO_2 emissions, energy consumption, $PM_{2.5}$ pollution, and energy efficiency	Renewable energy, institutional quality, carbon taxes, trade, and regulatory quality
16	Beka et al. (2024)	38 OECD and 5 Western Balkan countries	Comparative panel-data analysis using pooled ordinary least squares (OLS), fixed effects, random effects, and Hausman–Taylor IV	2010–2022	CO_2 emissions	Economic, financial, and institutional factors, particularly inflation, business freedom, and political stability

17	Bilalli et al. (2026)	Western Balkans	Dynamic panel-data analysis using fixed effects, random effects, generalized method of moments (GMM), two-stage least squares (2SLS)	2010–2022	CO_2 emissions	Domestic credit to the private sector, institutional quality, economic growth, foreign direct investment, and trade openness
18	Triarchi et al. (2025)	Bulgaria, Croatia, Greece, Romania, Slovenia, Albania, Bosnia and Herzegovina, North Macedonia, Montenegro, and Serbia	Comparative institutional analysis using inverse probability weighting (IPW) regression and average treatment effect estimation	1998–2022	CO_2 emissions	EU membership, institutional governance, economic development, and political and regulatory conditions
19	Pavlović & Trtica (2024)	Serbia	Scenario-based economic modelling; expenditure-side GDP projections	2023–2030	Real GDP growth under Emissions Trading System (ETS) implementation	Carbon pricing (ETS), electricity prices, household consumption, investments, GDP
20	Pettersson (2007)	Croatia, European Russia, Macedonia, Serbia, and Ukraine	Linear programming model based on the Regional Air Pollution Information and Simulation (RAINS) framework	Projection for 2020	Marginal cost of CO_2 emission reduction and transition to low-carbon or renewable electricity generation	Carbon pricing, CO_2 reduction target, renewable-energy availability, existing generation technologies, technology costs, and technological learning

No.	Author & Year	Main Focus	Key Findings	Main Contribution	DOI
13	Eicke et al. (2021)	Exposure and adaptive capacity to the EU CBAM	Long-term carbon lock-in and low trade diversification increase CBAM vulnerability; stronger MRV capacity supports adaptation	Provides a risk-index framework showing how economic dependence, emissions intensity, and institutional capacity influence third-country CBAM risks. Highlights the exposure of Bosnia and Herzegovina to CBAM because of its carbon intensity and institutional shortcomings	https://doi.org/10.1016/j.erss.2021.102240
14	Lobanov & Lobanova (2026)	Economic and trade effects of CBAM on carbon-intensive products	CBAM exposure varies according to export dependence, production structure, emissions intensity, and domestic carbon-pricing readiness; indicative costs	Product-level assessment of 35 carbon-intensive goods and cost estimates for Serbian exports	https://doi.org/10.4018/979-8-3373-8998-1.ch006

			are estimated for ten Serbian products		
15	Krasniqi & Gara (2025)	The role of state policies and institutional quality in preventing environmental pollution	Renewable energy ($B = -0.461$, $p < 0.001$), institutional quality ($B = -0.344$, $p < 0.01$), and carbon taxes ($B = -0.343$, $p < 0.001$) significantly reduce CO_2 emissions. Trade ($B = -0.481$, $p < 0.05$) and regulatory quality ($B = -0.0415$, $p < 0.05$) reduce energy consumption, while trade increases $PM_{2.5}$ pollution ($B = 0.261$, $p < 0.01$). Institutional quality improves energy efficiency ($B = 0.139$, $p < 0.05$)	Demonstrates that coordinated state interventions—including stronger institutions, carbon taxation, renewable-energy promotion, and regulatory improvements—can reduce emissions and improve environmental performance in the Western Balkans	https://doi.org/10.37410/EMERG.2025.3.05
16	Beka et al. (2024)	Comparative effects of economic, financial, and institutional dynamics on CO_2 emissions in OECD and Western Balkan countries	Inflation, business freedom, and political stability are identified as significant determinants of CO_2 emissions. The effects differ between OECD and Western Balkan countries, reflecting regional heterogeneity in economic, institutional, and environmental conditions	Highlights the significance of inflation, business freedom and political stability on CO_2 emissions, providing evidence from OECD and Western Balkan countries	https://doi.org/10.15388/Ekon.2024.103.3.1
17	Bilalli et al. (2026)	The impact of financial development and institutional quality on environmental quality	Domestic credit to the private sector and institutional quality has a positive and statistically significant effect on CO_2 emissions. Increased private-sector credit and improved institutional quality are associated with higher emissions	. Demonstrates that financial and institutional development do not automatically improve environmental quality and highlights the need to integrate environmental safeguards into financial and institutional policies	https://doi.org/10.17979/ejge.2026.15.1.12907
18	Triarchi et al. (2025)	The role of EU membership and institutional	EU membership is associated with stronger climate-mitigation	Provides comparative evidence that EU integration and	https://doi.org/10.1007/978-3-031-96428-2_10

		governance in advancing climate-change mitigation	performance and lower CO_2 emissions. EU member states demonstrate greater progress in reducing emissions, reflecting stricter environmental regulation and stronger governance frameworks	institutional governance influence climate-mitigation outcomes across Balkan economies, while highlighting the differences between EU and non-EU countries	
19	Pavlović & Trtica (2024)	Economic implications of carbon emissions trading in a coal-dependent economy	ETS implementation may slow GDP growth in the short term, while higher investment levels support economic recovery and stronger growth by 2030	Provides a scenario-based framework for assessing the macroeconomic effects of carbon pricing and investment responses during the energy transition	https://doi.org/10.1016/j.jup.2024.101855
20	Pettersson (2007)	Carbon pricing, renewable power generation and CO_2 reduction costs	The costs and technological pathways for reducing power-sector CO_2 emissions in Eastern European countries	Under an exogenously imposed 15% CO_2 reduction target, marginal abatement costs range from €4 to €90 per tonne of CO_2. The costs differ substantially across countries because of variations in renewable-energy potential, existing technologies, and production costs. Policy uncertainty and institutional barriers may also influence the allocation of climate-mitigation projects	https://doi.org/10.1016/j.enpol.2006.08.013

Note: A dash (–) indicates not applicable.

Research on topics such as the CBAM, carbon taxation, institutional quality, EU membership, and environmental regulation provide evidence suggesting that environmental policy is an important determinant of decarbonization outcomes and the mitigation of transition risks. The literature synthesis indicates that Western Balkan countries are vulnerable to EU climate policy instruments, particularly because of the carbon intensity and export orientation of certain economic sectors. Nevertheless, stronger institutional capacity, improved regulatory quality, and greater alignment with the EU Green Agenda may contribute to climate change mitigation efforts. The literature synthesis also reveals mixed findings regarding the role of institutions: while some studies suggest that stronger institutional quality is associated with lower greenhouse gas emissions, others indicate that institutional and financial development may stimulate economic activity and, consequently, increase pollution.

Overall, the reviewed evidence suggests that decarbonization in the Western Balkans is shaped by the interactions among energy structure, economic development, financial development, institutional quality, and external climate-policy pressures. The findings generally suggests that renewable energy use, energy-efficiency measures, carbon taxation, and environmental regulation can contribute to reductions in CO_2 emissions. However, the magnitude and direction of these effects may vary across countries depending on their economic, institutional,

and energy-related characteristics. In coal-dependent economies such as Kosovo, Serbia, and Bosnia and Herzegovina, carbon pricing and the EU CBAM may impose additional adjustment costs and increase economic transition risks. Regarding institutional quality and financial development, the evidence remains mixed. On the one hand, some studies indicate that improvements in institutional quality and regulatory frameworks contribute to better environmental outcomes. On the other hand, other studies suggest that institutional and financial development may stimulate economic growth and consequently increase emissions through higher investment, infrastructure development, energy consumption, and economic activity.

The literature further indicates that carbon intensity and decarbonization processes cannot be explained by a single factor. These processes depend on countries' energy profiles, their degree of dependence on coal, institutional capacity, the level of EU integration, access to green finance, and the environmental policies implemented in each country. Taken together, these findings highlight the need for a holistic and country-specific approach to the energy transition that accounts for differences in economic structure.

Theme 4, presented in Table 6, highlights the importance of the financial sector in shaping the green transition process and carbon emissions. The literature focusing on North Macedonia, Albania, and the broader region of the Western Balkans suggests that factors such as carbon taxes, green loans, banking sector development, and financial intermediation may influence CO_2 emissions reductions and transition risks. The reviewed studies suggest that green finance could facilitate emissions reductions by channeling credit towards sustainable investments. However, financial development more broadly may contribute to higher emissions by financing carbon-intensive activities. It is worth emphasizing that the key issue is not financial development itself, but rather the environmental orientation of financial activities.

Table 6. Theme 4—green finance, banking, and financial transition

No.	Author & Year	Country/Region	Study Design/Method	Period	Outcome Examined	Main Variables
21	Kovachevska-Stefanova & Gockov (2025)	North Macedonia	Banking-sector empirical analysis based on the construction and assessment of a loan carbon intensity indicator	–	Bank profitability, liquidity, and credit risk, measured through return on assets (ROA), liquidity coverage ratio (LCR), and non-performing loans (NPLs)	Loan carbon intensity, greenhouse-gas emissions, sectoral loan exposure, ROA, LCR, and NPLs
22	Lika & Lleshaj (2026)	Albania	Quantitative empirical study using before-and-after CO_2-emission measures and multiple linear regression	2019–2024	CO_2 savings following green investments	Loan amount, loan maturity, interest rate, total investment cost, and CO_2 emissions before and after investment
23	Turan & Toto (2025)	Western Balkans	Dynamic panel analysis using the augmented mean group (AMG) estimator, accounting for cross-sectional dependence and slope heterogeneity	2006–2019	Carbon emissions and environmental sustainability	Economic growth, squared economic growth, energy consumption, banking-sector development, financial development, and interaction or moderating effects

No.	Author & Year	Main Focus	Key Findings	Main Contribution	DOI
21	Kovachevska-Stefanova & Gockov (2025)	The effect of corporate-sector carbon intensity and transition risk	Total loan carbon intensity reduces long-term bank profitability, increases liquidity, and lowers credit risk. The	Develops the first loan carbon intensity indicator for North Macedonia and demonstrates that	https://doi.org/10.1080/13511610.2025.2507011

		on banking-sector performance	most carbon-intensive sectors are energy, transport and storage, manufacturing and construction, and agriculture. Based on the loan carbon intensity indicator, the sectors facing the greatest transition risk are water and waste management, energy, and agriculture and forestry. The effects of sectoral carbon footprints differ across economic sectors	corporate carbon exposure can affect bank profitability, liquidity, and credit risk, providing an initial framework for assessing climate-related transition risk in the banking sector	
22	Lika & Lleshaj (2026)	The effectiveness of credit-guarantee scheme-backed green loans in supporting CO_2 emission reductions among Albanian small and medium-sized enterprises (SMEs)	Higher loan amounts are significantly associated with greater CO_2 savings, suggesting that increased financial support strengthens the environmental effectiveness of green investments. Longer loan maturity has a positive but modest association with emission reductions. Higher interest rates are negatively associated with CO_2 savings, indicating that borrowing costs may discourage environmentally beneficial investments. Total investment cost is also negatively correlated with CO_2 savings, potentially reflecting inefficient allocation of investment resources	Provides firm-level evidence on how green-loan characteristics influence the environmental performance of SME investments. It also highlights the role of credit-guarantee schemes in reducing financing constraints and supporting low-carbon investments in Albania	https://doi.org/10.37394/232015.2026.22.14
23	Turan & Toto (2025)	The moderating role of banking-sector and financial development in the relationship between economic growth, energy consumption, and environmental degradation	The results support the validity of the Environmental Kuznets Curve hypothesis in the Western Balkans. Economic growth and energy consumption are significantly associated with environmental degradation. Banking-sector and financial development are inversely associated with carbon emissions, indicating that a stronger financial sector may mitigate some of the adverse environmental effects of economic growth	Provides regional evidence that banking and financial development can serve as moderating mechanisms in the growth–energy–emissions relationship. The study extends the EKC framework by incorporating the environmental role of the financial sector while using an estimator appropriate for heterogeneous and cross-sectionally dependent panels	https://doi.org/10.1016/B978-0-44-313776-1.00317-2

			and energy consumption			

Note: A dash (–) indicates not applicable.

This distinction is reflected in the reviewed literature, which suggests that green finance can support decarbonization through targeted credit allocation and effective management of climate-related risks. According to Turan & Toto (2025), financial development may help mitigate the environmental impacts of economic growth and energy consumption by directing credit towards more environmentally sustainable activities, while Lika & Lleshaj (2026) found that larger green loans are positively associated with reductions in CO_2 emissions; however, higher interest rates may weaken their environmental benefits. In contrast, Kovachevska-Stefanova & Gockov (2025) demonstrated that emissions-intensive sectors do not necessarily present the highest transition risks for banks, suggesting that carbon footprints and financial risks do not always coincide. Taken together, these findings indicate that the effectiveness of green finance depends on factors such as credit allocation, financing conditions, and climate-related risk assessment. However, the lack of harmonized regional data remains a significant limitation for cross-country comparisons in the Western Balkans.

Theme 5, presented in Table 7, highlights the contribution of various sectoral activities to carbon emissions in the Western Balkan region. Research on thermal power plants, construction, embodied carbon, real estate development, and urbanization indicates that carbon emissions are closely associated with material consumption, coal-fired electricity generation, and the expansion of construction activities. Findings from studies conducted in Serbia and Albania suggest that improvements in technical efficiency, cleaner production methods, and sustainable construction practices can help mitigate environmental pressures. However, the environmental benefits of such measures may remain limited without broader policy support and investment. Based on the reviewed literature covering Albania and the wider Western Balkan region, carbon emissions appear to be influenced by interactions among economic growth, construction activities, material consumption, urbanization, and environmental regulation. Vranjanac et al. (2018) provided a broader comparative perspective by examining CO_2 emission indicators across selected EU member states and Western Balkan countries. Their findings indicate that reductions in per capita CO_2 emissions were greater in the selected EU countries, including Austria and France, than in the Western Balkan countries examined.

Table 7. Theme 5—industrial sectors, construction, and eco-efficiency

No.	Author & Year	Country/ Region	Study Design/Method	Period	Outcome Examined	Main Variables
24	Vranjanac et al. (2018)	Germany, Austria, France, Serbia, North Macedonia, and Albania	Comparative environmental-indicator analysis based on the European Environment Agency methodology and relevant national methodologies	2006–2014	Outcome examined	National CO_2 emissions, population, GDP, level of economic development, and country group
25	Sefa et al. (2026)	Albania	Probabilistic process-based Life Cycle Assessment (LCA), structural inventory analysis, Monte Carlo simulation (N = 10,000), sensitivity analysis, scenario analysis, and typology-based scaling	Post-earthquake reconstruction period	Embodied carbon emissions from structural concrete at the material, building, and reconstruction-program levels	Cement content, concrete volume, structural design, building typology, and cement substitution.
26	Dorri & Shahini (2024)	Albania	Quantitative time-series analysis using the autoregressive distributed lag (ARDL) model and EViews 12 software	1995–2022	Carbon emissions associated with real estate and construction-sector development	Carbon emissions, building permits, GDP, and energy consumption patterns

27	Kiri & Sula (2023)	Albania	Quantitative time-series analysis using an ARDL model, ordinary least squares (OLS) estimation, Augmented Dickey–Fuller and Phillips–Perron unit-root tests, and diagnostic tests for autocorrelation and heteroskedasticity. The analysis was conducted in EViews using World Bank data	1990–2019	CO_2 emissions per capita, measured in metric tonnes, as an indicator of environmental pollution	CO_2 emissions per capita, urbanization measured as the percentage of the population living in urban areas, and GDP per capita

No.	Author & Year	Main Focus	Key Findings	Main Contribution	DOI/URL
24	Vranjanac et al. (2018)	Comparison of carbon-emission trends and eco-efficiency between selected EU and Western Balkan countries	Per-capita CO_2 emissions decreased by approximately 10% in Germany, 20% in Austria and France, 27% in Serbia, and 32% in North Macedonia, but increased by 53.8% in Albania. CO_2 emissions relative to GDP declined by 40% in Serbia and 46.2% in North Macedonia, while remaining broadly unchanged in Albania and the selected EU countries	Demonstrates substantial cross-country differences in emission trends and carbon efficiency, showing that declining emissions per unit of GDP do not necessarily follow the same pattern as changes in per-capita emissions.	https://www.researchgate.net/publication/326941178_Comparative_analysis_of_CO_2_emissions_indicators_in_EU_countries_and_western_Balkan_countries_-_Assessment_of_their_contribution_to_climate_change
25	Sefa et al. (2026)	Assessment of embodied carbon generated by reinforced-concrete residential reconstruction in Durrës, Albania	The volume-weighted emission intensity was 0.325 t CO_2-eq/m³. Cement accounted for 80%–85% of total embodied emissions. The reconstruction program of 100 residential buildings generated approximately 2.15×10^5 t CO_2-eq. Partial substitution of ordinary Portland cement could reduce emissions by 25%–30%	Provides a scalable, Albania-specific framework for assessing embodied carbon in post-earthquake reconstruction and demonstrates the importance of low-carbon cement, material efficiency, and early-stage structural design decisions.	https://doi.org/10.15157/IJITIS.2026.9.2.1107-1134

26	Dorri & Shahini (2024)	Examining the relationship between economic growth, real estate development, construction activity, and environmental sustainability in Albania	Construction activity and real estate development are positively associated with increased carbon emissions. The findings indicate that Albania's current construction-led development model diverges from sustainable practices. Greater construction activity, economic expansion, and energy consumption contribute to a higher environmental footprint	Provides one of the first empirical assessments of the environmental impact of Albania's real estate sector and quantifies the relationship between construction growth, economic indicators, energy use, and carbon emissions. It also supports policies promoting sustainable construction techniques and renewable energy integration	https://doi.org/10.3390/su16229780
27	Kiri & Sula (2023)	Examining the short- and long-term relationships between urbanization, economic growth, and carbon emissions in Albania	The authors report that urbanization has a negative but statistically insignificant relationship with CO_2 emissions: a 1% increase in urbanization is associated with an approximately 0.121% reduction in emissions, with a *p*-value of 0.0622. Current GDP per capita has a positive but statistically insignificant effect at the 5% level, while previous-period CO_2 emissions have a strong positive and statistically significant effect on current emissions. The results indicate that sustainable urban development requires balancing economic expansion with emission-reduction policies	Provides Albania-specific time-series evidence on the dynamic relationship between urbanization, economic growth, and environmental pollution. It emphasizes the need for sustainable urban infrastructure, renewable energy investment, appropriate urban planning, and consideration of emission-intensive industries in development policies	https://doi.org/10.36948/ijfmr.2023.v05i04.5645

In this context, environmental regulations, the adoption of cleaner technologies, improvements in energy

efficiency, and appropriate environmental policies can help mitigate the adverse environmental impacts of economic development. At the sectoral level, Sefa et al. (2026) highlighted the significant contribution of construction materials and design to carbon emissions, while Dorri & Shahini (2024) identified positive correlations among construction activity, energy consumption, and carbon emissions in Albania. These findings suggest that environmental pressures depend not only on the scale of construction activities but also on the materials and technologies employed. Furthermore, Kiri & Sula (2023) indicated that unplanned urbanization contributes to increased consumption of land, energy, and water resources, as well as higher levels of pollution and waste generation. Taken together, these findings suggest that carbon-related environmental challenges in Albania and the wider Western Balkan region are shaped by interconnected elements of construction, urbanization, technology, and institutional factors.

Table 8 presents Theme 6, which examines the interrelationship between decarbonization and sustainable development and green growth. Research on sustainable development efficiency, innovation systems, and environmental sustainability indicates that Western Balkan countries lag behind EU member states in terms of sustainable development. Innovation capacity, institutional readiness, and sustainability-oriented policies are identified as key factors that may support decarbonization and green growth. The findings suggest that the green transition should not be viewed solely as an environmental challenge, but rather as a broader development strategy encompassing innovation, governance, financing, and regional alignment with EU standards and requirements. Following the narrative synthesis of the six thematic areas, a VOSviewer keyword co-occurrence analysis was conducted to provide an integrated overview of the key concepts identified across the 30 studies included in the review. Figure 3 illustrates the thematic structure of research on carbon intensity in the Western Balkans and the relationships among its main research themes.

Table 8. Theme 6—green growth, sustainability, and innovation

No.	Author & Year	Country/Region	Study Design/Method	Period	Outcome Examined	Main Variables
28	Mimović et al. (2026)	Western Balkan countries and selected former socialist EU member states	Comparative quantitative analysis using data envelopment analysis with a slacks-based measure model (SBM–DEA), inverse DEA, composite and harmonized efficiency measures, and the simple additive weighting (SAW) method with Criteria Importance Through Intercriteria Correlation (CRITIC)-derived weights	2019	Efficiency and overall performance of sustainable development	CO_2 emissions, unemployment rate, energy consumption, GDP, and life expectancy at birth
29	Stanković & Radukić (2026)	Western Balkans	Comparative cross-country analysis using innovation, artificial intelligence preparedness, green-growth, and greenhouse-gas indicators.	2010–2023	Green growth, total greenhouse-gas emissions, and greenhouse-gas emissions relative to GDP	AI Preparedness Index, Eco-Innovation Index, green-growth indicators, total greenhouse gas (GHG) emissions, and GHG emissions per unit of GDP
30	Bytyqi et al. (2025)	Western Balkans	Quantitative panel-data analysis using a panel vector autoregression (panel VAR) model, unit-root tests, and	–	Dynamic relationship between environmental sustainability	CO_2 emissions and GDP per capita growth.

			Granger causality tests	and economic development	
No.	**Author & Year**	**Main Focus**	**Key Findings**	**Main Contribution**	**DOI**
28	Mimović et al. (2026)	Comparison of sustainable development efficiency between Western Balkan countries and former socialist countries that are members of the EU	EU member states generally achieved higher sustainable development efficiency and performance than Western Balkan countries. Most Western Balkan economies were inefficient, although Albania performed comparatively better. Bulgaria and, to a lesser extent, Slovakia were notable lower-performing EU exceptions. The SAW results were consistent with the SBM–DEA findings	Provides comparative evidence that EU membership and stronger institutional and policy frameworks are generally associated with better sustainability performance. It also identifies the efficiency gap between Western Balkan and EU economies by jointly considering environmental, economic, and social indicators	https://doi.org/10.1007/978-3-032-08592-4_6
29	Stanković & Radukić (2026)	Examining whether more developed innovation ecosystems and stronger AI preparedness accelerate decarbonization and green growth in the Western Balkans	Specific empirical results are not reported in the abstract. The study proposes to assess whether Western Balkan countries with more developed innovation ecosystems achieve stronger green growth and lower total and GDP-adjusted GHG emissions	Connects innovation ecosystems, eco-innovation, and AI preparedness with decarbonization performance and provides a basis for developing strategic priorities, convergence policies, and green-growth measures in the Western Balkans	https://doi.org/10.2478/eoik-2026-0008
30	Bytyqi et al. (2025)	Examining the dynamic interaction and causal relationship between economic growth and environmental sustainability in the Western Balkans	The study identifies significant bidirectional causality between CO_2 emissions and GDP per capita growth, indicating that economic development affects environmental quality while changes in emissions also influence economic performance	Provides regional econometric evidence on the interconnected nature of economic growth and CO_2 emissions and highlights the importance of renewable energy investment, energy efficiency, and regional cooperation for achieving sustainable economic development	https://doi.org/10.1007/978-3-031-73510-3_13

Note: A dash (–) indicates not applicable.

Figure 3 reveals that carbon emissions and economic growth constitute the central themes of the reviewed literature. These themes are closely linked to energy consumption, renewable energy and decarbonization, EU integration, climate change mitigation, carbon pricing, and the CBAM. Environmental sustainability, institutional quality, financial development, construction, green growth, and trade openness represent additional interconnected themes. Larger nodes indicate a higher frequency of term occurrence in the selected studies, while stronger links and shorter distances reflect greater thematic relatedness. While Figure 3 illustrates the overall network of thematic associations, Figure 4 provides a more detailed representation of the relationships among

individual studies, geographical contexts, investigated mechanisms, and reported findings. Collectively, the six thematic areas indicate that research on carbon intensity in the Western Balkans is multidimensional, characterized by strong interconnections among economic, energy-related, institutional, technological, and policy variables. To complement the narrative synthesis and visualize the relationships among the identified research dimensions, a VOSviewer network was constructed using manually coded data extracted from the 30 studies included in the PRISMA-guided systematic review, as detailed in Appendix A (Table A1). Figure 4 illustrates the relationships among the included studies, their geographical coverage, the key mechanisms examined, and the outcomes reported.

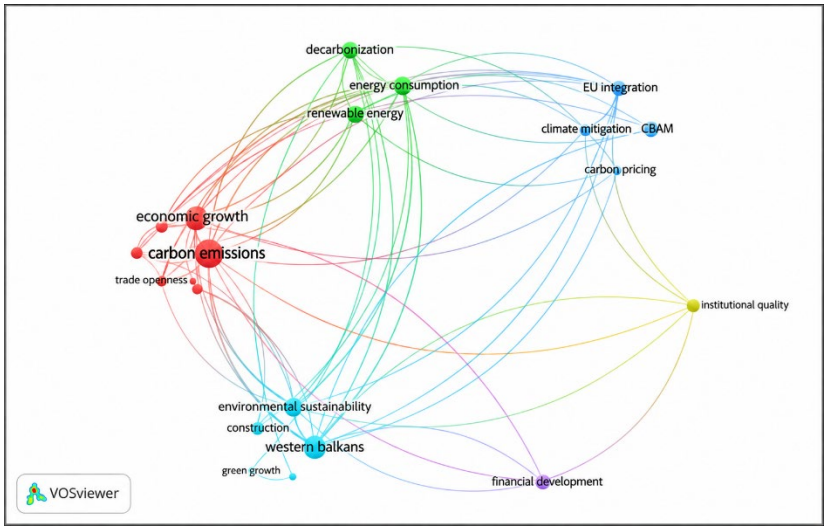


Figure 3. VOSviewer network visualization of thematic term co-occurrence in carbon-intensity research in the Western Balkans

Source: Authors' elaboration using VOSviewer, based on the manually coded terms extracted from the 30 studies included in the systematic review (see Appendix A, Table A1).

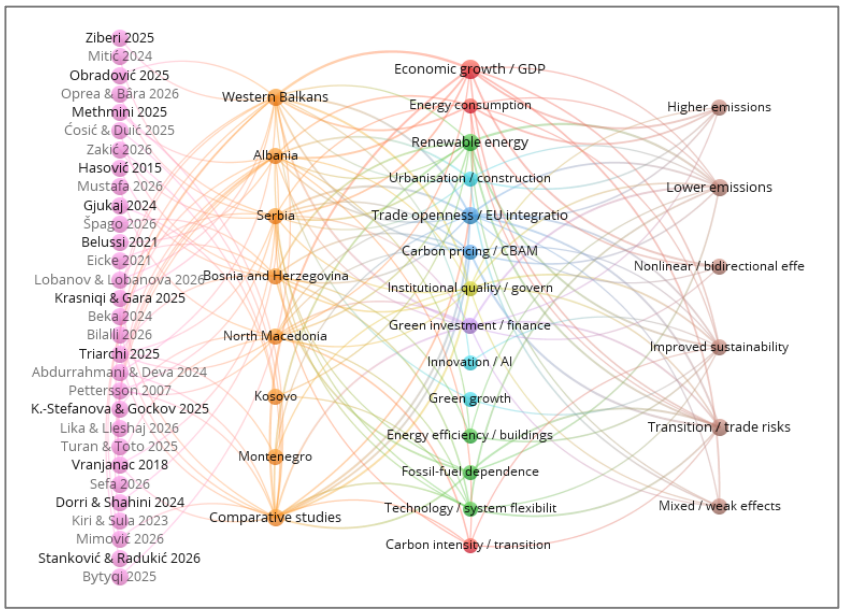


Figure 4. Network visualization of the relationships among included studies, geographical coverage, thematic mechanisms, and reported outcomes

Figure 4 illustrates the relationships among the 30 reviewed studies, their geographical scope, the thematic mechanisms examined, and the findings reported by the researchers. Based on the analysis, the main themes identified in the literature on carbon intensity in the Western Balkans can be summarized as follows: economic growth, energy consumption, renewable energy, trade and EU integration institutional quality, green investment, and technological transition. The reported outcomes encompass increases and decreases in emissions, as well as

broader sustainable development outcomes.

Notably, the six thematic dimensions indicate that CO_2 emissions and carbon intensity in the Western Balkans are shaped by interactions among economic, energy, institutional, financial, technological and sectoral determinants. Economic growth, industrialization, trade, investment, and urbanization may contribute to increased economic activity and energy demand, but their environmental impacts depend on the energy mix, energy efficiency, technology, and sectoral composition. Renewable energy, innovation, and technological advancement may contribute to reducing carbon intensity; however, institutional quality, environmental governance, climate policies, green finance, and effective policy implementation may determine the extent to which these environmental benefits are realized. Therefore, the findings suggest that the determinants identified across the six thematic dimensions should not be examined in isolation, as they are interconnected and may jointly influence environmental outcomes.

4. Discussion

The reviewed literature provides important insights into the interactions among economic growth, institutional effectiveness, innovation capacity, and environmental performance that shape the prospects for green growth in the Western Balkans. According to Mimović et al. (2026), Western Balkan countries are characterized by relatively low sustainability efficiency compared with former socialist EU member states. Stanković & Radukić (2026) highlighted the importance of eco-innovation adoption, artificial intelligence readiness, and supportive innovation ecosystems in accelerating decarbonization and green growth. Finally, Bytyqi et al. (2025) reported on the existence of bidirectional relation between CO_2 emissions and GDP per capita growth. Despite the general agreement that institutional and policy-related factors influence environmental performance in the Western Balkans, reported effects remains inconsistent. According to Krasniqi & Gara (2025), institutional quality, renewable energy use, and carbon taxation are negatively associated with CO_2 emissions, while regulatory quality contributes to reducing energy consumption. Similarly, Triarchi et al. (2025) found that EU accession and improved governance structures are associated with positive outcomes in climate change mitigation and emissions reductions.

However, Bilalli et al. (2026) reported a positive correlation between institutional quality and CO_2 emissions. One possible explanation is that improvements in institutional quality may stimulate investment, economic growth, and infrastructure development, thereby increasing energy demand before environmental benefits become apparent. This suggests that improvements in institutional quality do not necessarily lead to immediate reductions in emissions. Rather, the environmental effects of institutional development may depend on additional economic, structural, and policy-related factors. Differences in findings across these studies may also reflect variations in environmental outcome measures, model specifications, geographical coverage, and periods of analysis. While some studies examine absolute CO_2 emissions, others assess environmental quality or climate change mitigation performance. Therefore, the relationship between institutional quality and environmental outcomes should not be interpreted as uniformly positive or negative, but rather as context-dependent and influenced by the indicators and methodological approaches employed.

Based on reviewed literature, carbon pricing and the CBAM offer potential environmental benefits while also imposing significant economic adjustment costs on the region. Pavlović & Trtica (2024) demonstrated that introducing an ETS in Serbia's coal-dependent electricity sector may slow GDP growth in the short term, while increased investments could support economic recovery and stronger growth by 2030.

Furthermore, studies examining the CBAM provide additional insights into these challenges. Eicke et al. (2021) highlighted that vulnerability to the CBAM depends on ta country's level of exports to the EU, its emissions intensity, degree of trade diversification, carbon lock-in, and institutional capacity for monitoring, reporting, and verification. Lobanov & Lobanova (2026) further confirmed that the effects of the CBAM differ across Western Balkan countries and product categories depending on carbon intensity, exposure to the EU market, industrial structures, embedded emissions, and the development of carbon pricing policies. Taken together, these findings suggest that the consequences of carbon pricing and the CBAM should not be assessed solely in terms of emissions reductions. Their implications for production costs, export competitiveness, institutional readiness to implement carbon pricing mechanisms, and the adaptive capacity of firms and governments should also be considered when evaluating these policies.

Overall, the reviewed evidence suggests that renewable energy use, improvements in energy efficiency, and technological development are associated with reduced CO_2 emissions. However, the economic and technical feasibility of these approaches varies considerably across countries and sectors. Pettersson (2007) reported that marginal abatement costs per ton of CO_2, under a 15% emissions reduction target, ranged from €4 to €90 in the Eastern European countries examined. These differences reflect variations in the availability of renewable energy resources, electricity generation technologies, learning effects, and production costs.

These findings further support the argument that the region is characterized by considerable heterogeneity. Thus, countries with abundant renewable energy resources, diversified energy systems, and well-developed

institutions may be better positioned to achieve emissions reductions at relatively lower costs. In contrast, countries with coal-dependent economies may face higher adjustment costs, greater transition risks, and increased exposure to carbon pricing policies. Consequently, environmental outcomes may also differ depending on the policy instruments adopted and the initial technological conditions of each country. The relationship between financial development and environmental quality also remains mixed. According to Bilalli et al. (2026), domestic credit to the private sector is positively associated with CO_2 emissions, suggesting that financial development may support carbon-intensive investments, infrastructure expansion, and increased energy consumption when green lending requirements are insufficiently developed or enforced.

Beka et al. (2024) highlighted that financial, economic, and institutional variables are among the key determinants of CO_2 emissions, while emphasizing that their effects differ between OECD and Western Balkan countries. In other words, the environmental impact of financial development may depend more on the allocation of credit than on the total volume of lending. Loans that finance the conventional infrastructure and energy-intensive production may contribute to higher CO_2 emissions, whereas environmentally friendly lending, financing for renewable energy projects, and green lending requirements may support decarbonization. The main limitation of the existing literature is the limited number of studies that distinguish between conventional financial development and environmentally sustainable financing.

4.1 Research Gaps from Methodological and Regional Perspectives

Several key research gaps can be identified from the synthesis presented in this review. First, only a small proportion of the reviewed studies explicitly examines carbon intensity, while most focus on absolute CO_2 emissions or broader environmental indicators. Second, the literature exhibits uneven geographical coverage, with limited evidence on the impact of carbon taxation and the CBAM in Kosovo and Montenegro, while a greater number of studies focus on Serbia, Bosnia and Herzegovina, and regional panels. Third, existing research is predominantly based on aggregate national-level analyses, whereas sector-specific evidence remains limited, particularly in the energy, construction, and financial sectors. Fourth, relatively few studies jointly assess the environmental effectiveness and socioeconomic costs of carbon taxation, the CBAM, and other decarbonization policy instruments. Fifth, the existing literature does not adequately distinguish green finance from broader financial development.

The differences identified across the reviewed studies may be explained by several factors. First, the studies employ different environmental and economic outcome measures, such as carbon intensity, total CO_2 emissions, per capita emissions, environmental quality, transition risks, and abatement costs. Second, the studies apply a wide range of econometric approaches and analytical methods, including pooled ordinary least squares (OLS), fixed-effects models, random-effects models, generalized method of moments (GMM), two-stage least squares (2SLS), ARDL models, vector autoregression (VAR), VECM, inverse probability weighting (IPW), linear programming, and energy system simulations. These approaches rely on different assumptions and are designed to capture short-run, long-run, dynamic, or scenario-based relationships.

Third, Western Balkan countries differ considerably in terms of energy structure, institutional capacity, economic development, trade exposure, and progress towards EU integration. In particular, Serbia, Kosovo, and Bosnia and Herzegovina face greater exposure to coal-related transition risks. Furthermore, differences in renewable energy capacity, governance quality, and access to green finance may contribute to the heterogeneity observed across the reviewed studies. Therefore, the conflicting findings do not necessarily indicate methodological inconsistencies but rather reflect the heterogeneity of the existing literature and the diverse economic, institutional, and energy-related characteristics of the Western Balkan countries.

4.2 Policy Implications

The synthesis of the reviewed literature provides valuable insights into determinants of carbon intensity in the context of sustainable economic growth in the Western Balkans. However, these implications should be interpreted with caution, as the existing evidence remains fragmented across countries, methodologies, and research themes. At the regional level, the reviewed studies suggest that policy efforts should prioritize improvements in energy efficiency, renewable energy production, sustainable infrastructure development, low-carbon transportation, and stronger alignment with EU climate-policy instruments. Investment and capital formation should increasingly be directed towards low-carbon industries, as investments concentrated in fossil-fuel-dependent or energy-intensive sectors may contribute to higher carbon intensity.

At the same time, country-specific approaches remain essential, particularly given the differences among Western Balkan economies in terms of energy structures, institutional capacities, financial development, and exposure to carbon-intensive production. Accordingly, carbon pricing, green-finance instruments, renewable-energy priorities, and institutional reforms should be adapted to national conditions. The literature also highlights the need to reduce energy intensity and integrate urban and industrial policies with decarbonization strategies.

Overall, carbon intensity in the region is shaped by economic, energy, institutional, and structural factors, while comprehensive evidence from PRISMA-guided systematic reviews remains limited. This review contributes to addressing this research gap by systematically synthesizing and organizing the existing evidence into six analytical dimensions associated with carbon intensity and sustainable economic development in the Western Balkans.

4.3 Limitations and Future Research

This systematic literature review has several limitations. First, the literature search was conducted exclusively using the Scopus database because the authors had institutional access to Scopus but not to other databases, such as Web of Science. Consequently, relevant studies indexed exclusively in other databases may have been omitted. Second, the search strategy was based on the Scopus search strings reported in this study and was designed to identify relevant publications across six thematic areas. These search results formed the basis for the selection of the 30 studies included in the PRISMA-guided systematic review. Another limitation concerns the manual screening and classification process. The studies were selected following full-text assessment, and those considered relevant t to the predefined themes were included. No specialized digital or automated screening tools were used, which may have introduced some degree of subjectivity and affected the degree of the reproducibility of the study selection process. Data extraction was performed manually using an Excel spreadsheet. The extracted information included the main findings, geographical coverage, methodological characteristics, and thematic classification of the 30 studies. These data were subsequently used to construct the bibliometric network visualization. Therefore, the reliance on manual screening, data extraction, and thematic coding constitutes an additional limitation of the study. Future studies could build on the evidence base established by this review by expanding the literature search to additional academic databases and scholarly search platforms, such as Web of Science, ScienceDirect, and Google Scholar, thereby reducing database-related selection bias and identifying potentially relevant studies not indexed in Scopus. Future systematic reviews could employ specialized electronic screening and reference management tools to improve the transparency, consistency, and reproducibility of study selection and data extraction. Future empirical studies are needed to address the methodological heterogeneity identified in the reviewed literature and to examine Western Balkan economies that remain underrepresented in existing research. Future research could employ harmonized environmental indicators, consistent observation periods, and comparable econometric approaches to facilitate cross-country comparisons within the region. Further research could investigate sector-specific carbon intensity, the effects of renewable energy adoption, green innovation, environmental policies, institutional quality, and EU integration on decarbonization and sustainable economic growth. Studies employing longitudinal, panel-data, and comparative approaches covering all Western Balkan countries could provide deeper insights into the long-term dynamics of the green transition. Furthermore, future reviews could combine systematic literature review approaches with bibliometric analysis, meta-analysis, or qualitative content analysis to provide a more comprehensive assessment of the relationships among carbon emissions, carbon intensity, and sustainable economic growth.

5. Conclusion

This systematic literature review synthesized evidence from 30 studies identified through the Scopus database and selected according to predefined eligibility criteria, with the review process reported in accordance with the PRISMA 2020 guidelines. The reviewed evidence was classified into six thematic categories: (1) general CO_2 emissions, carbon intensity, and sustainable growth; (2) energy transition, renewable energy, and decarbonization; (3) environmental governance, institutions, and climate policy; (4) green finance, banking, and financial transition; (5) industrial sectors, construction, and eco-efficiency; and (6) green growth, sustainability, and innovation. Unlike approaches that treat these categories as independent research areas, the reviewed literature suggests that environmental outcomes in the Western Balkans are shaped by interactions among these dimensions. The synthesis further highlights that CO_2 emissions and carbon intensity should be distinguished as related but conceptually different indicators. While absolute CO_2 emissions represent the total quantity of carbon dioxide released, carbon intensity measures emissions relative to economic output, energy consumption, or production, depending on the indicator used. Consequently, improvements in carbon intensity do not necessarily translate into proportional reductions in absolute emissions, particularly when economic activity and energy demand continue to increase. Differences in the environmental indicators employed may therefore partly explain the inconsistent findings reported across studies.

The relationship between economic growth and environmental degradation also remains contested in the reviewed literature. Some studies provide evidence supporting an inverted-U-shaped EKC, while others identify a U-shaped relationship or report statistically insignificant effects. These divergences should not necessarily be interpreted as direct contradictions; rather, they may reflect differences in country samples, stages of economic development, energy structures, observation periods, environmental indicators, variable definitions, and methodological approaches. The integrated synthesis indicates that economic growth and industrial development

influence production levels and energy demand, whereas their environmental consequences depend on the energy mix, technology, sectoral composition, and resource-use efficiency. Institutional capacity, environmental governance, alignment with EU climate policies, green finance, and regulatory compliance may act as enabling or moderating factors. Effective institutions and consistent policy measures may enhance the environmental benefits of renewable energy deployment, technological innovation, and sustainable investment, whereas weak policy implementation may limit these benefits. Therefore, carbon intensity and CO_2 emissions in the Western Balkans should be understood as outcomes of interconnected economic, energy-related, institutional, financial, technological, and sectoral processes rather than as consequences of isolated determinants.

Furthermore, the review identifies significant geographical and methodological research gaps. Existing studies predominantly focus on absolute or per capita CO_2 emissions rather than carbon intensity, while relatively few provide comparative empirical evidence covering all Western Balkan countries. Uneven geographical coverage, differences in environmental indicators, observation periods, sectoral scope, and methodological approaches limit cross-study comparability. Future research should therefore prioritize harmonized environmental indicators, consistent observation periods, sector-specific analyses, and comparative approaches encompassing the entire Western Balkan region. Greater attention should also be devoted to examining the interrelationships among institutional development, green finance, innovation, energy transition, and sectoral transformation within the regional context.

Author Contributions

Conceptualization, B.Z., Y.P., N.K., S.T., R.B., and F.M.; methodology, B.Z.; software, R.B.; validation, S.T., N.K., and F.M.; formal analysis, N.K.; investigation, B.Z.; resources, Y.P.; data curation, S.T.; writing—original draft preparation, B.Z.; writing—review and editing, B.Z., Y.P., N.K., S.T., R.B., and F.M.; visualization, R.B.; supervision, N.K.; project administration, B.Z. and Y.P. All authors have read and agreed to the published version of the manuscript.

Data Availability

Not applicable.

Conflicts of Interest

The authors declare no conflicts of interest.

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Appendix

Table A1. Studies included in the systematic review and coding framework for the VOSviewer analysis

No.	Authors & Year	Country	Article Title	Source Title	Theme	DOI/URL
1	Ziberi et al. (2025)	Western Balkans	The EU Green Agenda legal framework and economic growth in developing countries: A panel EKC approach	Business: Theory and Practice	CO_2	https://doi.org/10.3846/btp.2025.23716
2	Mitić et al. (2024)	Serbia	An EKC-based modelling of CO_2 emissions, economic growth, electricity consumption and trade openness in Serbia	Environmental Science and Pollution Research	CO_2 emissions and environmental quality	https://doi.org/10.1007/s11356-023-31617-y
3	Obradović et al. (2025)	Western Balkans	Energy, urbanization and carbon footprint: Evidence from Western Balkan countries	Urban Science	CO_2 emissions and environmental quality	https://doi.org/10.3390/urbansci9040119
4	Oprea & Bâra (2026)	Serbia	Can Serbia reconcile economic growth with sustainability in a policy space shaped by	Journal of Business Economics and Management	Economic growth, CO_2 emissions, and renewable-energy consumption	https://doi.org/10.3846/jbem.2026.26926

			contrasting regional norms?			
5	Methmini et al. (2025)	Twenty high-emitting countries across five continents, including Albania	Impact of economic growth, energy consumption, and trade openness on carbon emissions: evidence from the top 20 emitting nations	Climate and Development	GDP, energy consumption, and trade openness	https://doi.org/10.1080/17565529.2024.2375737
6	Ćosić & Duić (2025)	Croatia and Bosnia and Herzegovina	Impact of the EU carbon border adjustment mechanism and variable renewable energy integration on fossil-based energy systems	Energy Conversion and Management	CO_2 intensity, fossil-fuel dependence, electricity trade, renewable-energy curtailment, and system flexibility	https://doi.org/101008.10.1016/j.ecmx.2025.101008
7	Zakić et al. (2026)	Serbia, with comparative reference to China and the European Union	Sino-Serbian renewable energy cooperation: A path towards economic growth	In Energy for Sustainable Development: Capitalizing on the Transformative Potential of Energy to Promote Economic Growth and Improve Livelihoods	Renewable-energy transition readiness and economic-growth potential	https://doi.org/10.1007/978-3-032-04071-8_16
8	Hasović et al. (2015)	Bosnia and Herzegovina	Impact of new power investments up to year 2020 on the energy system of Bosnia and Herzegovina.	Thermal science	CO_2 emissions, renewable-energy target compliance, electricity production, and energy-system performance	https://doi.org/10.2298/TSCI150105042H
9	Mustafa et al. (2026)	North Macedonia	Investments in renewable energy and decarbonization of North Macedonia	In International Scientific Conference on Business and Economics	CO_2 emissions, economic growth, and inflation	https://doi.org/10.1007/978-3-032-04214-9_18
10	Gjukaj et al. (2024)	Kosovo, with a case study of the Sibovc coal basin	Towards a sustainable energy future in Kosovo–Balancing development, environmental concerns and renewable solutions	Ecological Engineering & Environmental Technology	Solar-energy generation potential, projected CO_2-emission reduction, and environmental feasibility	https://doi.org/10.12912/27197050/189603
11	Špago et al. (2026)	Bosnia and Herzegovina; five cities from central, northern, and southern regions	Techno-economic analysis of residential solar collectors for carbon dioxide emission reduction in Bosnia and Herzegovina	In International Conference "New Technologies, Development and Applications"	CO_2-emission reduction, solar-energy coverage, household energy-cost savings, and investment payback period	https://doi.org/10.1007/978-3-032-30107-9_13
12	Belussi et al. (2021)	Albania; public-school	Achieving near zero energy	E3S Web of Conferences	Primary energy performance and	https://doi.org/10.1051/e3sconf/202131202005

		case study in Tirana	building in Albania: An approach for the retrofit of a public-school building		feasibility of achieving the nearly Zero Energy Building standard	
13	Eicke et al. (2021)	Global and Bosnia and Herzegovina case	Pulling up the carbon ladder? Decarbonization, dependence, and third-country risks from the European carbon border adjustment mechanism	Energy Research & Social Science	Carbon border adjustment mechanism (CBAM) within the European Green Deal,	https://doi.org/10.1016/j.erss.2021.102240
14	Lobanov & Lobanova (2026)	Western Balkans, with detailed estimates for Serbia	CBAM and the Western Balkans: Assessing impacts on production and exports of 35 carbon-intensive goods	In The Ripple Effects of Environmental Policies: Economic, Financial, and Institutional Perspectives	Assessing Impacts on Production and Exports of 35 Carbon-Intensive Goods CBAM-related production, export exposure, and compliance costs	https://doi.org/10.4018/979-8-3373-8998-1.ch006
15	Krasniqi & Gara (2025)	Western Balkans	Role of the state in the prevention of environmental pollution: Empirical evidence for the countries of the Western Balkans	Energy. Environment. Efficiency. Resources. Globalization	CO_2 emissions, energy consumption, $PM_{2.5}$ pollution, and energy efficiency	https://doi.org/10.37410/EMERG.2025.3.05
16	Beka et al. (2024)	38 OECD and 5 Western Balkan countries	Assessing the role of economic, financial, and institutional dynamics on CO_2 emissions: Comparative analysis of OECD and Western Balkan regions	Ekonomika	CO_2 emissions	https://doi.org/10.15388/Ekon.2024.103.3.1
17	Bilalli et al. (2026)	Western Balkans	The impact of financial development and institutional quality on environmental quality in the Western Balkans	European Journal of Government and Economics	CO_2 emissions	https://doi.org/10.17979/ejge.2026.15.1.12907
18	Triarchi et al. (2025)	Bulgaria, Croatia, Greece, Romania, Slovenia, Albania, Bosnia and Herzegovina, North Macedonia, Montenegro, and Serbia	EU membership and institutional governance in advancing climate change mitigation: A comparative analysis of the Balkan economies	In Building Economic Resilience: Strategies for Sustainable Growth and Competitiveness	CO_2 emissions	https://doi.org/10.1007/978-3-031-96428-2_10
19	Pavlović & Trtica (2024)	Serbia	Projecting short to medium-term effects of carbon	Utilities Policy	Carbon pricing (ETS), electricity prices, GDP	https://doi.org/10.1016/j.jup.2024.101855

			emissions trading on Serbia's GDP		growth, household consumption and investments	
20	Pettersson (2007)	Croatia, European Russia, Macedonia, Serbia, and Ukraine	Carbon pricing and the diffusion of renewable power generation in Eastern Europe: A linear programming approach	Energy Policy	Carbon pricing, CO_2 reduction target, renewable-energy availability, existing generation technologies, technology costs, and technological learning	https://doi.org/10.1016/j.enpol.2006.08.013
21	Kovachevska-Stefanova & Gockov (2025)	North Macedonia	Investigating the impact of loan carbon intensity indicator on selected banking sector indicators in North Macedonia	Innovation: The European Journal of Social Science Research	Bank profitability, liquidity, and credit risk, measured through return on assets (ROA), liquidity coverage ratio (LCR), and non-performing loans (NPLs)	https://doi.org/10.1080/13511610.2025.2507011
22	Lika & Lleshaj (2026)	Albania	Green Loans and Their Impact on CO_2 Reduction: Evidence from credit guarantee scheme-backed loans in Albania	WSEAS Transactions on Environment and Development,	CO_2 savings following green investments	https://doi.org/10.37394/232015.2026.22.14
23	Turan & Toto (2025)	Western Balkans	The moderating effects of the banking and financial sector in achieving sustainable development through economic growth, energy consumption, and decarbonization: Dynamic panel estimate for the western Balkans	Encyclopedia of Monetary Policy, Financial Markets and Banking Volume 4	Carbon emissions and environmental sustainability	https://doi.org/10.1016/B978-0-44-313776-1.00317-2
24	Vranjanac et al. (2018)	Germany, Austria, France, Serbia, North Macedonia, and Albania	Comparative analysis of CO_2 emissions indicators in EU countries and western Balkan countries—Assessment of their contribution to climate change	Journal of Environmental Protection and Ecology	National CO_2 emissions, population, GDP, level of economic development, and country group	https://www.researchgate.net/publication/326941178_Comparative_analysis_of_CO2_emissions_indicators_in_EU_countries_and_western_Balkan_countries_-_Assessment_of_their_contribution_to_climate_change
25	Sefa et al. (2026)	Albania	Embodied carbon impacts of post-earthquake reconstruction in reinforced concrete buildings: A life-cycle assessment	International Journal of Innovative Technology and Interdisciplinary Sciences	Embodied carbon emissions from structural concrete at the material, building, and reconstruction-program levels	https://doi.org/10.15157/IJITIS.2026.9.2.1107-1134
26	Dorri & Shahini (2024)	Albania	Balancing economic and construction	Sustainability	Carbon emissions, building permits, GDP, and energy	https://doi.org/10.3390/su16229780

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27	Kiri & Sula (2023)	Albania	Urban development and impact on sustainable economic development: The case of Albania	International Journal for Multidisciplinary Research	Carbon dioxide emissions per capita, measured in metric tonnes, as an indicator of environmental pollution	https://doi.org/10.36948/ijfmr.2023.v05i04.5645
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30	Bytyqi et al. (2025)	Western Balkans	Environmental sustainability and economic development: A panel VAR analysis of the dynamic relationship in Western Balkan	In International Scientific Conference on Business and Economics	CO_2 emissions and GDP per capita growth	https://doi.org/10.1007/978-3-031-73510-3_13